E173017546_1

The equation $2x^4 + xy + y^3 + y^2 + 1 = 0$ has no integer solutions

Size $h = 49$, length $l = 12$

Theorem 1. *The equation*

$$2x^4 + xy + y^3 + y^2 + 1 = 0 \tag{1}$$

has no integer solutions.

The proof uses the cubic Hilbert reciprocity law over the field $K = \mathbb{Q}(\omega)$, where $\omega^2 + \omega + 1 = 0$. To a hypothetical integer solution we attach the values of 59 auxiliary polynomials $H_0, \ldots, H_{58}$ with coefficients in $\mathbb{Z}[\omega]$, constants $c_0, \ldots, c_{58} \in \mathbb{Z}[\omega]$, and a sum B of cubic Hilbert symbols of these values, recorded by a graph with 352 weighted edges. By the reciprocity law, the local contributions B_p of all primes p add up to 0 modulo 3. We show that $B_p = 0$ for $p \notin S$, compute B_p for the primes $p \in S$, and obtain a contradiction. Two standard facts from class field theory are quoted without proof: the explicit formula for the cubic Hilbert symbol at the places not above 3, and the Hilbert reciprocity law. Apart from these, the proof uses polynomial identities, which can be checked by expanding both sides, and finite computations with residues.

1 Cubic Hilbert symbols over $\mathbb{Q}(\omega)$

The field. Let $K = \mathbb{Q}(\omega)$, where $\omega^2 + \omega + 1 = 0$. Its ring of integers is $\mathbb{Z}[\omega]$, and every element of K can be written uniquely as $A + B\omega$ with $A, B \in \mathbb{Q}$. We write

$$\langle A, B\rangle = A + B\omega,$$

also when A and B are polynomials in x and y. Then $\langle A, B\rangle\langle C, D\rangle = \langle AC - BD,\ AD + BC - BD\rangle$, and the norm is $N(\langle A, B\rangle) = A^2 - AB + B^2$. In particular, a value $\langle A, B\rangle$ with $A, B \in \mathbb{Z}$ is zero if and only if $A = B = 0$.

Places. The finite places of K correspond to the non-zero prime ideals of $\mathbb{Z}[\omega]$. A prime $p \equiv 2 \pmod 3$ is inert: there is one place above p, with uniformizer p and residue field $\mathbb{Z}[\omega]/p\mathbb{Z}[\omega]$ with p^2 elements. If $p \equiv 1 \pmod 3$, the polynomial $t^2 + t + 1$ has two roots r modulo p, and for each of them there is a place v_r above p, at which ω is the root of $t^2 + t + 1$ in $\mathbb{Z}_p$ congruent to r modulo p; the completion is $\mathbb{Q}_p$, the uniformizer is p, the residue field is $\mathbb{F}_p$, and ω reduces to r. The prime 3 is ramified: $\pi = 1 - \omega$ generates the unique prime ideal above 3, $\pi^2 = -3\omega$, and $\mathbb{Z}[\omega]/\pi\mathbb{Z}[\omega] = \mathbb{F}_3$, where $\langle A, B\rangle$ reduces to $A + B$. The only infinite place of K is complex. For a finite place v we write K_v for the completion and $v(a)$ for the normalized valuation, and we call a a *unit* at v if $v(a) = 0$.

Hilbert symbols. For a place v of K and $a, b \in K_v^\times$, the cubic Hilbert symbol $(a, b)_v$ is a cube root of unity, see [1, Chapter V, §3]. We write $(a, b)_v = \omega^{\beta_v(a,b)}$ with $\beta_v(a, b) \in \mathbb{Z}/3\mathbb{Z}$, and we use the following three facts [1, Chapter V, §3 and Chapter VI, §8].

(H1) $\beta_v(a, bc) = \beta_v(a, b) + \beta_v(a, c)$, $\beta_v(a, b) = -\beta_v(b, a)$, and $\beta_v(a, c^3) = 0$. In particular, $\beta_v(a, b)$ depends only on the classes of a and b in $K_v^\times/K_v^{\times 3}$.

(H2) Let v be a finite place not above 3, let q be the number of elements of its residue field, and let ϖ be a uniformizer at v. Write $a = \varpi^e u$ and $b = \varpi^f w$ with units u and w, and define $\chi(u) \in \mathbb{Z}/3\mathbb{Z}$ by $\bar{u}^{(q-1)/3} = \bar{\omega}^{\chi(u)}$, where the bar denotes the reduction to the residue field. Then $\beta_v(a,b) = e\,\chi(w) - f\,\chi(u)$. In particular, $\beta_v(a,b) = 0$ if a and b are both units.

(H3) (Hilbert reciprocity.) For $a, b \in K^\times$, $\beta_v(a,b) = 0$ for all but finitely many places v, and $\sum_v \beta_v(a,b) = 0$, where the sum is over all places of K. At the complex place, $\beta_v(a,b) = 0$.

The general formula in (H2) contains an additional term $ef\chi(-1)$, which vanishes because $-1 = (-1)^3$ is a cube. Formula (H2) fixes the normalization of the symbols; with the opposite sign convention all values β_v change sign, and the proof below is the same.

Classes at the places not above 3. Let $p \neq 3$ be a prime and v a place above p. For $a = p^e u \in K_v^\times$ with a unit u, put $c_v(a) = (e \bmod 3,\ \chi(u)) \in \mathbb{F}_3^2$. By (H2) and (H1), c_v is additive, $c_v(a)$ determines the class of a modulo cubes, and $\beta_v(a,b) = e\chi(w) - f\chi(u)$ for $c_v(a) = (e, \chi(u))$ and $c_v(b) = (f, \chi(w))$. We put $c_p(a) = c_v(a) \in \mathbb{F}_3^2$ if $p \equiv 2 \pmod 3$, and $c_p(a) = (c_{v_r}(a), c_{v_{r'}}(a)) \in \mathbb{F}_3^4$ if $p \equiv 1 \pmod 3$, where $r < r'$ are the roots of $t^2 + t + 1$ in $\{1, \ldots, p-1\}$; accordingly, β_p is the sum of the forms β_v over the places v above p. For $p \equiv 2 \pmod 3$ and a rational integer u prime to p, we have $\bar{u} \in \mathbb{F}_p$, so that $\bar{u}^{(p^2-1)/3} = (\bar{u}^{p-1})^{(p+1)/3} = 1$ and $\chi(u) = 0$.

The place above 3.

Lemma 2. *Let v be the place above 3. Every non-zero element of K_v is equal to $\pi^a \omega^b 2^c (1+3\omega)^d$ times a cube, for unique $a, b, c, d \in \mathbb{Z}/3\mathbb{Z}$, and every unit congruent to 1 modulo 9 is a cube. Put $c_3(A) = (a,b,c,d) \in \mathbb{F}_3^4$. Then c_3 is additive, and $\beta_v(A,B) = c_3(A)^\mathsf{T} M_3\, c_3(B)$, where*

$$M_3 = \begin{pmatrix} 0 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \pmod 3.$$

Proof. For $\alpha \in \mathbb{Z}[\omega]$, the polynomial $g(s) = s + 3s^2 + 3s^3 - \alpha$ satisfies $g'(s) \equiv 1 \pmod \pi$, so by Hensel's lemma it has a root s in the completion of $\mathbb{Z}[\omega]$, and then $(1+3s)^3 = 1 + 9(s + 3s^2 + 3s^3) = 1 + 9\alpha$. Hence every unit congruent to 1 modulo 9 is a cube in K_v, and the class of a unit modulo cubes depends only on its residue modulo 9. There are 54 units in $\mathbb{Z}[\omega]/(9)$, and a direct enumeration shows that the 27 products $\omega^b 2^c (1+3\omega)^d$ with $b, c, d \in \{0,1,2\}$ are pairwise distinct modulo 9, also after multiplication by $-1 = (-1)^3$. Hence they represent all 27 classes of units modulo cubes. Since π^3 is a cube times a unit, the exponent of π is the valuation modulo 3, and the first statement follows.

By (H1), it remains to compute β_v on the 16 ordered pairs of the elements π, ω, 2 and $1+3\omega$. Their norms are 3, 1, 4 and 7, hence, for each such pair, all the symbols at the places not above 2, 3 and 7 vanish by (H2), and, by (H3), β_v at the place above 3 is minus the sum of the values at the place above 2 and at the two places above 7. These places correspond to $\bar{\omega} = 2$ and $\bar{\omega} = 4$ in $\mathbb{F}_7$. The classes (e, χ) of the four elements at these places are

	2	$\bar{\omega} = 2$	$\bar{\omega} = 4$
π	$(0,2)$	$(0,0)$	$(0,2)$
ω	$(0,1)$	$(0,2)$	$(0,2)$
2	$(1,0)$	$(0,2)$	$(0,1)$
$1+3\omega$	$(0,2)$	$(1,0)$	$(0,0)$

where, at the places above 7, $\chi(u)$ is defined by $\bar{u}^2 = \bar{\omega}^{\chi(u)}$ in $\mathbb{F}_7$, and 7 is a uniformizer. Formula (H2) and the sum over these three places give M_3. $\square$

We write β_3 for the form of Lemma 2. Since $3 = -\omega^2\pi^2$ and $4 = 2^2$, we have $c_3(3) = (2, 2, 0, 0)$ and $c_3(4) = (0, 0, 2, 0)$.

Residue classes. The local computations use the following description of the classes of the values of a polynomial on a residue class.

Lemma 3. *Let p be a prime, $k \geq 1$, and let $z \in \mathbb{Z}[\omega]$ be non-zero with $z \equiv \zeta \pmod{p^k}$, where $\zeta = \langle A, B\rangle$ with $0 \leq A, B < p^k$. Then $c_p(z) \in b + D$ for the vector b and the subspace D given as follows.*

(i) $p \equiv 2 \pmod 3$. If $\zeta \neq 0$, then $b = c_p(\zeta)$ and $D = 0$. If $\zeta = 0$, then $b = 0$, and D is $\mathbb{F}_3^2$, or the span of $(1, 0)$ if $z \in \mathbb{Z}$.

(ii) $p \equiv 1 \pmod 3$. For each root r, let ρ be the root of $t^2 + t + 1$ modulo p^k congruent to r. If $A + B\rho \not\equiv 0 \pmod{p^k}$, the two coordinates of $c_{v_r}(z)$ are those of $c_{v_r}(\zeta)$; otherwise these two coordinates are arbitrary.

(iii) $p = 3$, $z \notin \mathbb{Z}$. If $\zeta \neq 0$, let $a = v(\zeta) < 2k$ be its valuation at π and $m = 2k - a$. Then $b = c_3(\zeta)$, and $D = 0$ if $m \geq 4$, while $D = D_m$ if $m \leq 3$, where D_1 is spanned by $(0, 1, 0, 0)$, $(0, 0, 1, 0)$ and $(0, 0, 0, 1)$, D_2 by $(0, 0, 1, 0)$ and $(0, 0, 0, 1)$, and D_3 by $(0, 0, 1, 1)$. If $\zeta = 0$, then $b = 0$ and $D = \mathbb{F}_3^4$.

(iv) $p = 3$, $z \in \mathbb{Z}$, so that $B = 0$. If $3^k \nmid A$, let $j = v_3(A)$. Then $b = c_3(A)$ and $D = 0$ if $k - j \geq 2$, while $b = j\, c_3(3)$ and D is spanned by $c_3(4)$ if $k - j = 1$. If $3^k \mid A$, then $b = 0$ and D is spanned by $c_3(3)$ and $c_3(4)$.

Proof. (i) If $\zeta \neq 0$, then $e = v(z) = v(\zeta) < k$ and $z/p^e \equiv \zeta/p^e \pmod p$, so that χ is determined. If $z \in \mathbb{Z}$, its unit part is a rational integer, and $\chi = 0$. (ii) At v_r, z and ζ are the p-adic numbers $A' + B'\omega$ and $A + B\omega$ with $\omega \equiv \rho \pmod{p^k}$, and the argument of (i) applies. (iii) If $\zeta \neq 0$, then $v(z) = a$, and $z/\pi^a \equiv \zeta/\pi^a \pmod{\pi^m}$. By Lemma 2, the class of a unit depends only on its residue modulo $9 = \omega\pi^4$; the units congruent to 1 modulo π^m form a subgroup, and a direct enumeration of their residues modulo 9 shows that their classes form D_m. (iv) If $3^k \nmid A$, then $z = 3^j u$ with a rational unit $u \equiv A/3^j \pmod{3^{k-j}}$. The class of a rational unit depends only on its residue modulo 9, the units ± 1 are cubes, and the rational units congruent to 1 modulo 3 are 4^s times a unit congruent to 1 modulo 9; hence the classes are as stated, and if $3^k \mid A$ the valuation j is also unknown. □

Lemma 4. *Let p be a prime, and consider vertices i with vectors b_i, subspaces D_i and vectors γ_i, and a set of edges (i, j) with $i < j$ and weights $w_{ij} \in \{1, 2\}$. Put $\rho_{ij} = w_{ij}$ and $\rho_{ji} = -w_{ij}$ for the edges, and $\sigma_i = \gamma_i + \sum_j \rho_{ij} b_j$. Suppose that $\beta_p(d, \sigma_i) = 0$ for every i and every $d \in D_i$, and that $\beta_p(d, d') = 0$ for every edge (i, j) and all $d \in D_i$, $d' \in D_j$. Then*

$$\sum_{(i,j)} w_{ij}\beta_p(c_i, c_j) + \sum_i \beta_p(c_i, \gamma_i) = \sum_{(i,j)} w_{ij}\beta_p(b_i, b_j) + \sum_i \beta_p(b_i, \gamma_i)$$

whenever $c_i \in b_i + D_i$ for every i. By bilinearity, it suffices to check the hypotheses for d and d' in spanning sets of D_i and D_j.

Proof. Write $c_i = b_i + d_i$ with $d_i \in D_i$. As β_p is bilinear and $\beta_p(u, v) = -\beta_p(v, u)$, the left side equals the right side plus $\sum_i \beta_p(d_i, \sigma_i) + \sum_{(i,j)} w_{ij}\beta_p(d_i, d_j)$, and both sums vanish. □

2 The graph

Put $F(x,y) = 2x^4 + xy + y^3 + y^2 + 1$, so that (1) is the equation $F(x,y) = 0$. The certificate consists of 59 polynomials H_i with coefficients in $\mathbb{Z}[\omega]$ and constants $c_i \in \mathbb{Z}[\omega]$, listed in Table 1, and of 352 *edges* (i,j) with $i < j$ and weights $w_{ij} \in \{1,2\}$, listed below. For an edge (i,j) we put $\rho_{ij} = w_{ij}$ and $\rho_{ji} = -w_{ij} \bmod 3$, and $\rho_{ij} = 0$ if i and j are not joined by an edge. The edges $(i,j;w_{ij})$ are $(0,7;1)$, $(0,8;1)$, $(0,10;2)$, $(0,24;2)$, $(0,29;2)$, $(0,30;2)$, $(0,43;2)$, $(0,47;2)$, $(0,48;2)$, $(0,54;1)$, $(0,57;1)$, $(1,24;2)$, $(1,32;2)$, $(1,35;1)$, $(1,36;2)$, $(1,47;2)$, $(1,48;2)$, $(1,51;1)$, $(1,53;1)$, $(1,54;1)$, $(2,20;1)$, $(2,21;1)$, $(2,22;1)$, $(2,23;2)$, $(2,24;1)$, $(2,33;1)$, $(2,40;2)$, $(2,49;2)$, $(2,53;1)$, $(2,54;1)$, $(3,5;1)$, $(3,6;2)$, $(3,7;1)$, $(3,10;2)$, $(3,14;1)$, $(3,17;1)$, $(3,18;1)$, $(3,19;1)$, $(3,22;1)$, $(3,26;2)$, $(3,40;2)$, $(3,47;2)$, $(3,48;1)$, $(3,49;2)$, $(3,54;1)$, $(3,58;1)$, $(4,6;1)$, $(4,8;2)$, $(4,18;2)$, $(4,19;1)$, $(4,22;1)$, $(4,27;2)$, $(4,36;2)$, $(4,38;2)$, $(4,40;2)$, $(4,55;1)$, $(5,14;1)$, $(5,17;2)$, $(5,22;1)$, $(5,26;2)$, $(5,27;1)$, $(5,29;2)$, $(5,31;2)$, $(5,32;2)$, $(5,36;1)$, $(5,37;2)$, $(5,45;1)$, $(5,49;1)$, $(6,12;1)$, $(6,16;1)$, $(6,18;1)$, $(6,27;2)$, $(6,32;1)$, $(6,34;2)$, $(6,40;2)$, $(6,43;2)$, $(6,47;1)$, $(6,48;2)$, $(6,54;2)$, $(6,55;1)$, $(6,58;2)$, $(7,14;1)$, $(7,16;2)$, $(7,25;1)$, $(7,29;2)$, $(7,30;2)$, $(7,37;2)$, $(7,40;2)$, $(7,48;2)$, $(7,54;1)$, $(7,56;2)$, $(7,57;1)$, $(8,12;2)$, $(8,15;2)$, $(8,17;2)$, $(8,19;2)$, $(8,23;2)$, $(8,29;1)$, $(8,32;2)$, $(8,46;1)$, $(8,54;2)$, $(8,55;1)$, $(8,56;1)$, $(9,17;2)$, $(9,29;1)$, $(9,31;2)$, $(9,33;2)$, $(9,50;2)$, $(9,52;1)$, $(9,56;2)$, $(10,12;2)$, $(10,14;2)$, $(10,17;2)$, $(10,20;2)$, $(10,22;1)$, $(10,24;2)$, $(10,29;2)$, $(10,30;2)$, $(10,32;1)$, $(10,33;2)$, $(10,35;1)$, $(10,42;1)$, $(10,43;2)$, $(10,44;2)$, $(10,47;1)$, $(11,15;2)$, $(11,20;2)$, $(11,22;2)$, $(11,24;2)$, $(11,30;2)$, $(11,31;1)$, $(11,34;1)$, $(11,42;1)$, $(11,43;2)$, $(11,44;2)$, $(12,17;2)$, $(12,23;2)$, $(12,29;2)$, $(12,33;2)$, $(12,34;1)$, $(12,44;1)$, $(12,49;2)$, $(12,54;2)$, $(13,21;2)$, $(13,46;2)$, $(13,51;1)$, $(13,52;1)$, $(14,16;2)$, $(14,22;2)$, $(14,24;2)$, $(14,26;1)$, $(14,27;2)$, $(14,29;2)$, $(14,43;2)$, $(14,47;2)$, $(14,51;1)$, $(14,53;1)$, $(14,54;2)$, $(15,28;2)$, $(15,42;2)$, $(15,58;1)$, $(16,24;1)$, $(16,25;2)$, $(16,30;1)$, $(16,34;1)$, $(16,38;2)$, $(16,40;2)$, $(16,43;1)$, $(16,51;2)$, $(16,53;1)$, $(17,22;2)$, $(17,27;1)$, $(17,28;1)$, $(17,32;2)$, $(17,34;2)$, $(17,36;2)$, $(17,39;1)$, $(17,41;2)$, $(17,47;2)$, $(17,50;2)$, $(17,51;2)$, $(18,27;1)$, $(18,34;1)$, $(18,41;2)$, $(18,49;1)$, $(18,52;1)$, $(18,54;1)$, $(18,56;2)$, $(19,28;2)$, $(19,36;2)$, $(19,50;1)$, $(19,52;1)$, $(20,22;2)$, $(20,23;2)$, $(20,27;2)$, $(20,31;1)$, $(20,33;1)$, $(20,35;2)$, $(20,36;1)$, $(20,40;2)$, $(20,52;1)$, $(20,55;1)$, $(20,57;1)$, $(20,58;2)$, $(21,22;2)$, $(21,25;1)$, $(21,32;1)$, $(21,34;2)$, $(21,38;1)$, $(21,43;2)$, $(21,48;1)$, $(21,52;2)$, $(21,54;1)$, $(21,58;1)$, $(22,26;1)$, $(22,40;1)$, $(22,48;2)$, $(22,50;2)$, $(22,52;2)$, $(23,25;1)$, $(23,26;1)$, $(23,36;2)$, $(23,49;1)$, $(23,50;2)$, $(23,51;1)$, $(23,56;2)$, $(24,28;2)$, $(24,35;2)$, $(24,38;2)$, $(24,40;1)$, $(24,42;2)$, $(24,44;1)$, $(24,48;2)$, $(24,52;2)$, $(25,26;1)$, $(25,27;2)$, $(25,28;2)$, $(25,30;1)$, $(25,35;1)$, $(25,38;1)$, $(25,42;1)$, $(25,44;1)$, $(25,49;1)$, $(25,57;2)$, $(26,27;1)$, $(26,46;1)$, $(26,49;2)$, $(26,50;1)$, $(26,56;1)$, $(27,28;2)$, $(27,37;2)$, $(27,41;1)$, $(27,43;2)$, $(27,52;1)$, $(27,54;1)$, $(27,55;2)$, $(28,35;2)$, $(28,37;1)$, $(28,38;1)$, $(28,40;1)$, $(28,41;1)$, $(28,46;1)$, $(28,48;2)$, $(28,49;1)$, $(28,50;1)$, $(28,51;2)$, $(28,58;2)$, $(29,32;1)$, $(29,33;2)$, $(29,40;2)$, $(29,45;1)$, $(29,47;1)$, $(29,50;2)$, $(29,54;2)$, $(30,31;1)$, $(30,44;1)$, $(30,45;1)$, $(30,46;2)$, $(30,47;1)$, $(30,48;2)$, $(30,50;2)$, $(30,51;1)$, $(30,52;1)$, $(31,40;2)$, $(31,46;1)$, $(31,47;1)$, $(31,54;1)$, $(31,58;1)$, $(32,40;1)$, $(32,42;1)$, $(32,43;1)$, $(33,43;2)$, $(33,44;2)$, $(33,49;2)$, $(34,35;2)$, $(34,43;2)$, $(34,46;2)$, $(34,47;1)$, $(34,48;2)$, $(34,49;1)$, $(34,54;1)$, $(35,44;2)$, $(35,45;2)$, $(35,49;1)$, $(35,52;2)$, $(35,53;2)$, $(35,54;2)$, $(36,40;1)$, $(36,46;2)$, $(36,47;2)$, $(36,52;1)$, $(36,55;1)$, $(37,51;1)$, $(37,56;2)$, $(37,58;2)$, $(38,39;2)$, $(38,56;1)$, $(38,58;2)$, $(39,52;1)$, $(39,56;2)$, $(40,42;1)$, $(40,48;1)$, $(40,51;2)$, $(40,56;2)$, $(41,50;2)$, $(41,58;1)$, $(42,47;2)$, $(42,49;1)$, $(43,45;2)$, $(43,52;2)$, $(43,57;2)$, $(44,50;1)$, $(44,53;1)$, $(44,57;1)$, $(45,47;1)$, $(45,48;1)$, $(46,47;1)$, $(46,48;2)$, $(47,49;1)$, $(47,51;1)$, $(47,56;2)$, $(48,51;1)$, $(48,55;2)$, $(48,57;2)$, $(49,54;1)$, $(50,52;1)$, $(50,54;1)$, $(50,56;1)$, $(50,58;1)$, $(52,54;2)$, $(52,56;1)$, $(54,58;2)$. Let (ξ,η) denote (x,y) or (y,x). We call a vertex i *linear* if $H_i = \varepsilon_i(\lambda_i\eta - s_i(\xi))$ for a unit ε_i of $\mathbb{Z}[\omega]$, an element $\lambda_i \in \mathbb{Z}[\omega]$ whose norm is a power of 3, and a polynomial s_i with coefficients in $\mathbb{Z}[\omega]$; the last column of Table 1 gives $\lambda_i\eta = s_i(\xi)$ for the linear vertices, where $\lambda_i = 1$ if no factor is shown. For a linear vertex

i, we put $f_i(t) = \lambda_i^d F$ and $g_{ij}(t) = \lambda_i^{d_j} H_j$ evaluated at $\xi = t$, $\eta = s_i(t)/\lambda_i$, where d and d_j are the degrees of F and H_j in η; these are polynomials in t with coefficients in $\mathbb{Z}[\omega]$. For every linear vertex i, Table 2 gives an integer $D_i \geq 1$ and a polynomial $w_i(t)$ with coefficients in $\mathbb{Z}[\omega]$ such that

$$D_i^3 \, \lambda_i^{e_i} c_i \prod_j g_{ij}^{\rho_{ij}} - w_i^3 = f_i Q_i \tag{2}$$

for a polynomial $Q_i(t)$ with coefficients in $\mathbb{Z}[\omega]$, where $\rho_{ij} \in \{0, 1, 2\}$; the table also gives an integer $e_i \geq 0$ with $e_i + \sum_j \rho_{ij} d_j \equiv 0 \pmod 3$, and $e_i = 0$ if $\lambda_i = 1$; Q_i is the quotient of the division of the left side by f_i, and the remainder is zero. For every other vertex i, Table 2 gives an integer $D_i \geq 1$ and polynomials W_i, A_i and B_i in x and y with coefficients in $\mathbb{Z}[\omega]$ such that

$$D_i^3 \, c_i \prod_j H_j^{\rho_{ij}} - W_i^3 = A_i F + B_i H_i. \tag{3}$$

For the edges (i, j) listed in Table 3, the table gives a linear endpoint $k \in \{i, j\}$ and the factorization of the norm of the resultant $R_{ij} = \mathrm{Res}_t(f_k, g_{kl})$, where $\{k, l\} = \{i, j\}$; it is non-zero. These identities can be checked by expanding both sides and by polynomial division, using $\omega^2 = -1 - \omega$.

3 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1), so that $F(x, y) = 0$; from now on the polynomials denote their values at (x, y), which lie in $\mathbb{Z}[\omega]$.

Step 1: the values are non-zero. For $i \in \{0, 1, 2, 4, 5, 6, 8, 9, 10, 11, 12, 13, 14, 15, 17, 18, 19, 23, 24, 25, 26, 27, 28, 29, 31, 32, 33, 35, 36, 37, 38, 39, 41, 42, 45, 46, 47, 48, 49, 50, 51, 53, 54, 55, 56, 58\}$, there is no residue pair (a, b) modulo 2 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod 2$, as one checks for the 2^2 residue pairs; hence $H_i \neq 0$. For $i \in \{3, 21, 22, 30, 34, 57\}$, there is no residue pair (a, b) modulo 3 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod 3$, as one checks for the 3^2 residue pairs; hence $H_i \neq 0$. For $i \in \{7, 16, 43, 44\}$, there is no residue pair (a, b) modulo 4 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod 4$, as one checks for the 4^2 residue pairs; hence $H_i \neq 0$. For $i \in \{20, 40, 52\}$, there is no residue pair (a, b) modulo 5 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod 5$, as one checks for the 5^2 residue pairs; hence $H_i \neq 0$. Also, all c_i are non-zero.

Step 2: reciprocity. For every place v of K, put

$$B_v = \sum_{(i,j)} w_{ij} \, \beta_v(H_i, H_j) + \sum_i \beta_v(H_i, c_i) \in \mathbb{Z}/3\mathbb{Z}, \tag{4}$$

the first sum over the edges, and, for a rational prime p, put $B_p = \sum_{v|p} B_v$. Applying (H3) to every pair (H_i, H_j) with an edge, multiplied by w_{ij}, and to every pair (H_i, c_i), and adding, we find that $B_v = 0$ for all but finitely many v and $\sum_p B_p = 0$, as the complex place contributes 0. By (H1) and the definition of c_p, $B_p = \sum_{(i,j)} w_{ij} \beta_p(c_p(H_i), c_p(H_j)) + \sum_i \beta_p(c_p(H_i), c_p(c_i))$, where β_p is the form of Lemma 2 if $p = 3$.

Step 3: the primes outside S. Let $S = \{2, 3, 5, 7, 11, 13, 19\}$. It contains 3 and the primes dividing the norms of the constants c_i, of the D_i, of the R_{ij}. Let $p \notin S$, and let v be a place above p, with residue field k_v; then all these numbers are units at v. First, no edge (i, j) has both values divisible by v. Indeed, suppose that v divides H_i and H_j. If R_{ij} is defined, let k be the linear endpoint used for it and $\{k, l\} = \{i, j\}$, and put $t = \xi$. As $H_k = \varepsilon_k(\lambda_k \eta - s_k(\xi))$ and λ_k is a unit at v, we have $\eta \equiv s_k(t)/\lambda_k$ modulo v, hence $f_k(t) \equiv \lambda_k^d F(x, y) = 0$ and $g_{kl}(t) \equiv \lambda_k^{d_l} H_l \equiv 0$ modulo v. The resultant can be written as $R_{ij} = A f_k + C g_{kl}$ with polynomials A and C with

coefficients in $\mathbb{Z}[\omega]$, so v divides R_{ij}, a contradiction. By (H2), a term of (4) in which all arguments are units vanishes. Now let v divide H_i. Collecting the terms of (4) which contain H_i, and using (H1) and $\rho_{ji} = -w_{ij}$ for the edges (j, i) with $j < i$, we obtain $\beta_v(H_i, \Pi_i)$ with $\Pi_i = c_i \prod_j H_j^{\rho_{ij}}$, which is a unit. We claim that $D_i^3 \Pi_i \equiv W^3$ modulo v for some $W \in K$ integral at v. If i is linear, put $t = \xi$; then $g_{ij}(t) \equiv \lambda_i^{d_j} H_j$ and $f_i(t) \equiv 0$ modulo v as above, and (2) gives $D_i^3 \lambda_i^{E_i} \Pi_i \equiv w_i(t)^3$ modulo v, where $E_i = e_i + \sum_j \rho_{ij} d_j$ is divisible by 3; this gives the claim with $W = w_i(t)/\lambda_i^{E_i/3}$. Otherwise, (3) gives the claim with $W = W_i(x, y)$. Hence the reduction of Π_i is a non-zero cube in k_v, $\chi(\Pi_i) = 0$, and $\beta_v(H_i, \Pi_i) = v(H_i)\chi(\Pi_i) = 0$ by (H2). Hence $B_v = 0$, and $B_p = 0$.

Step 4: the primes in S. Let $p \in S$. Here 2 is inert, 5 is inert, the places above 7 are v_2 and v_4, 11 is inert, the places above 13 are v_3 and v_9, and the places above 19 are v_7 and v_{11}. Suppose that $x \equiv a$ and $y \equiv b \pmod{p^k}$. Then $H_i \equiv H_i(a, b) \pmod{p^k}$, and Lemma 3, applied with ζ the residue of $H_i(a, b)$, gives a vector b_i and a subspace D_i with $c_p(H_i) \in b_i + D_i$; when H_i has rational coefficients, $H_i \in \mathbb{Z}$, and we use this in (i) and (iv). With $\gamma_i = c_p(c_i)$, if the hypotheses of Lemma 4 hold, then B_p is determined by a, b and k. Starting from the solutions of (1) modulo p, we refine a residue class $x \equiv a$, $y \equiv b \pmod{p^k}$ to the classes modulo p^{k+1} that contain solutions of (1) modulo p^{k+1}, until these hypotheses hold, or until no solution remains. Table 4 lists the resulting terminal classes and the value of B_p in each; every integer solution lies in one of them. The result is $B_2 = 1$, $B_3 \in \{0, 2\}$, $B_5 = 0$, $B_7 = 1$, $B_{11} = 0$, $B_{13} = 0$, $B_{19} = 0$.

Step 5: contradiction. By Steps 2 and 3, $\sum_{p \in S} B_p = 0$. But by Step 4, $\sum_{p \in S} B_p \in \{1, 2\}$ in $\mathbb{Z}/3\mathbb{Z}$. This contradiction proves the theorem. $\square$

References

[1] J. Neukirch, *Algebraic Number Theory*, Grundlehren der mathematischen Wissenschaften 322, Springer, Berlin, 1999.

Table 1: The polynomials H_i, the constants c_i and, for the linear vertices, $\eta = r_i(\xi)$. Here $\langle A, B\rangle = A + B\omega$.

i	H_i	c_i	chart
0	$\langle -y, 0\rangle$	$\langle 2, 0\rangle$	$y = \langle 0, 0\rangle$
1	$\langle -y, 1\rangle$	$\langle -4, -4\rangle$	$y = \langle 0, 1\rangle$
2	$\langle -y, -1\rangle$	$\langle 0, -4\rangle$	$y = \langle 0, -1\rangle$
3	$\langle -y-1, 0\rangle$	$\langle 50, -50\rangle$	$y = \langle -1, 0\rangle$
4	$\langle -y-1, -1\rangle$	$\langle 2, -2\rangle$	$y = \langle -1, -1\rangle$
5	$\langle x+1, 1\rangle$	$\langle -1, -2\rangle$	$x = \langle -1, -1\rangle$
6	$\langle x, -1\rangle$	$\langle 1, 0\rangle$	$x = \langle 0, 1\rangle$
7	$\langle x-1, 0\rangle$	$\langle 0, -6\rangle$	$x = \langle 1, 0\rangle$
8	$\langle x-1, -1\rangle$	$\langle 4, 0\rangle$	$x = \langle 1, 1\rangle$
9	$\langle x-1, 1\rangle$	$\langle -50, -50\rangle$	$x = \langle 1, -1\rangle$
10	$\langle 2x+1, 0\rangle$	$\langle 50, 0\rangle$	–
11	$\langle 2x-1, 0\rangle$	$\langle 0, -12\rangle$	–
12	$\langle -2x-y, -x\rangle$	$\langle -1, 2\rangle$	$y = \langle -2x, -x\rangle$
13	$\langle -y+2, -x+1\rangle$	$\langle -100, -50\rangle$	$y = \langle 2, -x+1\rangle$
14	$\langle -2x-y, -x+1\rangle$	$\langle 0, -3\rangle$	$y = \langle -2x, -x+1\rangle$
15	$\langle -2x-y-1, -x\rangle$	$\langle 24, 9\rangle$	$y = \langle -2x-1, -x\rangle$
16	$\langle 2x-y-1, x-1\rangle$	$\langle 1, 0\rangle$	$y = \langle 2x-1, x-1\rangle$
17	$\langle x-y-1, -x-1\rangle$	$\langle -12, -8\rangle$	$y = \langle x-1, -x-1\rangle$
18	$\langle -x-y-1, x-1\rangle$	$\langle 1, 0\rangle$	$y = \langle -x-1, x-1\rangle$
19	$\langle -2x-2y-1, -x-1\rangle$	$\langle 25, 125\rangle$	$\langle 1, -1\rangle\, x = \langle -1, 2y\rangle$
20	$\langle x-y, 0\rangle$	$\langle 2, -2\rangle$	$y = \langle x, 0\rangle$
21	$\langle x+y, 0\rangle$	$\langle 0, -3\rangle$	$y = \langle -x, 0\rangle$
22	$\langle x+1, -y+1\rangle$	$\langle 1, 0\rangle$	$y = \langle -x, -x-1\rangle$
23	$\langle x+y+1, 1\rangle$	$\langle 1, 3\rangle$	$y = \langle -x-1, -1\rangle$
24	$\langle x+y+1, y+1\rangle$	$\langle 0, -2\rangle$	$y = \langle -1, x\rangle$
25	$\langle x-y+1, -y\rangle$	$\langle 1, 0\rangle$	$y = \langle 0, -x-1\rangle$
26	$\langle x-y+1, 0\rangle$	$\langle 25, -25\rangle$	$y = \langle x+1, 0\rangle$
27	$\langle x+y, 1\rangle$	$\langle 0, -3\rangle$	$y = \langle -x, -1\rangle$
28	$\langle x, -y-1\rangle$	$\langle 0, -4\rangle$	$y = \langle -x-1, -x\rangle$
29	$\langle x+y, -1\rangle$	$\langle 1, 0\rangle$	$y = \langle -x, 1\rangle$
30	$\langle x+y, y-1\rangle$	$\langle 0, -3\rangle$	$y = \langle 1, x+1\rangle$
31	$\langle x-y-1, -y\rangle$	$\langle 12, -20\rangle$	$y = \langle 0, -x+1\rangle$
32	$\langle x-y-1, 0\rangle$	$\langle -3, 0\rangle$	$y = \langle x-1, 0\rangle$
33	$\langle x+y-1, y\rangle$	$\langle 0, -3\rangle$	$y = \langle 0, x-1\rangle$
34	$\langle x-1, -y-1\rangle$	$\langle -3, -2\rangle$	$y = \langle -x, -x+1\rangle$
35	$\langle x+y-1, -1\rangle$	$\langle 1, 0\rangle$	$y = \langle -x+1, 1\rangle$
36	$\langle x+y-1, y-1\rangle$	$\langle 98, 0\rangle$	$y = \langle 1, x\rangle$
37	$\langle x-y-1, 1\rangle$	$\langle -15, 0\rangle$	$y = \langle x-1, 1\rangle$
38	$\langle x-y-2, -1\rangle$	$\langle 0, -15\rangle$	$y = \langle x-2, -1\rangle$
39	$\langle x-2y-1, -y-2\rangle$	$\langle -10, -10\rangle$	$x = \langle 2y+1, y+2\rangle$
40	$\langle 2x-y+1, 0\rangle$	$\langle -75, -50\rangle$	$y = \langle 2x+1, 0\rangle$
41	$\langle 2x+y, 1\rangle$	$\langle 25, -25\rangle$	$y = \langle -2x, -1\rangle$
42	$\langle 2x-y, -y-1\rangle$	$\langle 0, -15\rangle$	$y = \langle -1, -2x-1\rangle$
43	$\langle 3x-y, -2y\rangle$	$\langle -6, -16\rangle$	$\langle 1, -1\rangle\, y = \langle -3x, -3x\rangle$
44	$\langle 3x+y, 2y\rangle$	$\langle 125, 200\rangle$	$\langle 1, -1\rangle\, y = \langle 3x, 3x\rangle$
45	$\langle x^2-y+1, x^2+1\rangle$	$\langle -2, -1\rangle$	$y = \langle x^2+1, x^2+1\rangle$
46	$\langle x^2-y+1, x^2\rangle$	$\langle 1, 0\rangle$	$y = \langle x^2+1, x^2\rangle$
47	$\langle x^2+x-y, x^2+x\rangle$	$\langle 3, -5\rangle$	$y = \langle x^2+x, x^2+x\rangle$
48	$\langle x^2-y, x\rangle$	$\langle 275, -50\rangle$	$y = \langle x^2, x\rangle$
49	$\langle -y, -x^2-x\rangle$	$\langle -5, 3\rangle$	$y = \langle 0, -x^2-x\rangle$
50	$\langle x^2-x-y, 0\rangle$	$\langle 25, -25\rangle$	$y = \langle x^2-x, 0\rangle$
51	$\langle -x-y, x^2\rangle$	$\langle -3, -3\rangle$	$y = \langle -x, x^2\rangle$
52	$\langle -x^2-x-y+1, -x^2-x\rangle$	$\langle 0, -3\rangle$	$y = \langle -x^2-x+1, -x^2-x\rangle$
53	$\langle x^2-y, x^2+x+1\rangle$	$\langle 0, -15\rangle$	$y = \langle x^2, x^2+x+1\rangle$

i	H_i	c_i	chart
54	$\langle -x-y-1, x^2-1\rangle$	$\langle -3,-3\rangle$	$y=\langle -x-1, x^2-1\rangle$
55	$\langle -x-2y, x^2-1\rangle$	$\langle 1,0\rangle$	–
56	$\langle x^2-x-2y-1, 0\rangle$	$\langle -2,-1\rangle$	–
57	$\langle -2x^2-3y+1, -x^2-1\rangle$	$\langle 1,0\rangle$	$3y=\langle -2x^2+1, -x^2-1\rangle$
58	$\langle x^2-2x-3y-1, 2x^2-x+1\rangle$	$\langle -50,25\rangle$	$3y=\langle x^2-2x-1, 2x^2-x+1\rangle$

Table 2: The data of the identities (2) and (3); the entry $j{:}e$ means $\rho_{ij}=e\neq 0$.

$i=0,\ D_i=4,\ \rho_{ij}$: 7:1, 8:1, 10:2, 24:2, 29:2, 30:2, 43:2, 47:2, 48:2, 54:1, 57:1

$$w_i \quad \langle 12t^3-12t-18,\ 12t^3+18t^2+6t\rangle$$

$i=1,\ D_i=4,\ \rho_{ij}$: 24:2, 32:2, 35:1, 36:2, 47:2, 48:2, 51:1, 53:1, 54:1

$$w_i \quad \langle -34t^2+40t-18,\ 36t^3-44t^2+20t+6\rangle$$

$i=2,\ D_i=2,\ \rho_{ij}$: 20:1, 21:1, 22:1, 23:2, 24:1, 33:1, 40:2, 49:2, 53:1, 54:1

$$w_i \quad \langle 6t^3-4t^2-4t+2,\ 16t^3+14t^2+6t+6\rangle$$

$i=3,\ D_i=8,\ \rho_{ij}$: 5:1, 6:2, 7:1, 10:2, 14:1, 17:1, 18:1, 19:1, 22:1, 26:2, 40:2, 47:2, 48:1, 49:2, 54:1, 58:1

$$w_i \quad \langle 340t^3-120t-180,\ -340t^3+120t+180\rangle$$

$i=4,\ D_i=38,\ \rho_{ij}$: 6:1, 8:2, 18:2, 19:1, 22:1, 27:2, 36:2, 38:2, 40:2, 55:1

$$w_i \quad \langle -68t^3+88t^2-305t+204,\ 134t^3-388t^2+311t-3\rangle$$

$i=5,\ D_i=4,\ \rho_{ij}$: 3:2, 14:1, 17:2, 22:1, 26:2, 27:1, 29:2, 31:2, 32:2, 36:1, 37:2, 45:1, 49:1

$$w_i \quad \langle -48t^2-192t-216,\ 12t^2-60t-108\rangle$$

$i=6,\ D_i=8,\ \rho_{ij}$: 3:1, 4:2, 12:1, 16:1, 18:1, 27:2, 32:1, 34:2, 40:2, 43:2, 47:1, 48:2, 54:2, 55:1, 58:2

$$w_i \quad \langle -216t^2-576t+3168,\ -1512t^2-1152t+2232\rangle$$

$i=7,\ D_i=4,\ \rho_{ij}$: 0:2, 3:2, 14:1, 16:2, 25:1, 29:2, 30:2, 37:2, 40:2, 48:2, 54:1, 56:2, 57:1

$$w_i \quad \langle 480t^2+672t-288,\ 168t^2-168t-1008\rangle$$

$i=8,\ D_i=4,\ \rho_{ij}$: 0:2, 4:1, 12:2, 15:2, 17:2, 19:2, 23:2, 29:1, 32:2, 46:1, 54:2, 55:1, 56:1

$$w_i \quad \langle 408t^2-232t-280,\ 8t^2+456t-776\rangle$$

$i=9,\ D_i=4,\ \rho_{ij}$: 17:2, 29:1, 31:2, 33:2, 50:2, 52:1, 56:2

$$w_i \quad \langle 80t^2+360t+880,\ -120t^2+80t-560\rangle$$

$i=10,\ D_i=256,\ \rho_{ij}$: 0:1, 3:1, 12:2, 14:2, 17:2, 20:2, 22:1, 24:2, 29:2, 30:2, 32:1, 33:2, 35:1, 42:1, 43:2, 44:2, 47:1

$$W \quad \langle -20748y^2+9408y-13797,\ -17124y^2+8754y-7506\rangle$$

$$\begin{aligned}
A \quad \langle & -4586471424000x^{21}y^3+4892236185600x^{21}y^2+9478707609600x^{21}y \\
& -15288238080000x^{20}y^4+11924825702400x^{20}y^3+38832124723200x^{20}y^2 \\
& +11619060940800x^{20}y-14472865382400x^{19}y^5+6421059993600x^{19}y^4 \\
& +81129583411200x^{19}y^3+42093615513600x^{19}y^2-18142042521600x^{19}y \\
& +11822904115200x^{18}y^6+14676708556800x^{18}y^5+51521362329600x^{18}y^4 \\
& +42195537100800x^{18}y^3-58655873433600x^{18}y^2-52183852646400x^{18}y \\
& +42331432550400x^{17}y^7+47104760217600x^{17}y^6-30185776742400x^{17}y^5 \\
& -118195067289600x^{17}y^4-187416811929600x^{17}y^3-142044718694400x^{17}y^2 \\
& -37863869644800x^{17}y+41176321228800x^{16}y^8+49194152755200x^{16}y^7 \\
& -113251870310400x^{16}y^6-374239081267200x^{16}y^5-353752842240000x^{16}y^4 \\
& -176596136755200x^{16}y^3-21488467968000x^{16}y^2+54324205977600x^{16}y \\
& +5220650188800x^{15}y^9-14925850214400x^{15}y^8-229578375168000x^{15}y^7 \\
& -390829650739200x^{15}y^6-456826709606400x^{15}y^5-131694015283200x^{15}y^4 \\
& +188835220684800x^{15}y^3+154258322227200x^{15}y^2+109158019891200x^{15}y \\
& -31686288998400x^{14}y^{10}-120493965312000x^{14}y^9-263025642700800x^{14}y^8 \\
& -215683065446400x^{14}y^7-107855688499200x^{14}y^6+108424750694400x^{14}y^5 \\
& +306498030796800x^{14}y^4+230121956966400x^{14}y^3+154088452915200x^{14}y^2 \\
& +55649186611200x^{14}y-42795742003200x^{13}y^{11}-167438293401600x^{13}y^{10} \\
& -132295163904000x^{13}y^9+68355411148800x^{13}y^8+489119809536000x^{13}y^7 \\
& +375332850892800x^{13}y^6+185573729894400x^{13}y^5+193421692108800x^{13}y^4 \\
& -55793575526400x^{13}y^3-78479622144000x^{13}y^2-52617019392000x^{13}y \\
& -29334542745600x^{12}y^{12}-89785368576000x^{12}y^{11}+75386113228800x^{12}y^{10} \\
& +477966001766400x^{12}y^9+845689651200000x^{12}y^8+546179855155200x^{12}y^7 \\
& +29152876953600x^{12}y^6-171086708736000x^{12}y^5-287570342707200x^{12}y^4 \\
& -213890944204800x^{12}y^3-193047979622400x^{12}y^2-109107059097600x^{12}y \\
& -5315022028800x^{11}y^{13}+43711148851200x^{11}y^{12}+272838426624000x^{11}y^{11}
\end{aligned}$$

$$
\begin{aligned}
&+726303611289600x^{11}y^{10}+826959672115200x^{11}y^{9}+507763438387200x^{11}y^{8}\\
&+39512073830400x^{11}y^{7}-221331220070400x^{11}y^{6}+62785113292800x^{11}y^{5}\\
&+105420895027200x^{11}y^{4}-44008892006400x^{11}y^{3}-69790806835200x^{11}y^{2}\\
&-60694305177600x^{11}y+13936833331200x^{10}y^{14}+126073228492800x^{10}y^{13}\\
&+363408968908800x^{10}y^{12}+623899783987200x^{10}y^{11}+546825830400000x^{10}y^{10}\\
&+274290809241600x^{10}y^{9}-162563516006400x^{10}y^{8}-315465714892800x^{10}y^{7}\\
&-152493568819200x^{10}y^{6}-59018261299200x^{10}y^{5}+134624260915200x^{10}y^{4}\\
&+145348676812800x^{10}y^{3}+58829989478400x^{10}y^{2}+7440275865600x^{10}y\\
&+19372651315200x^{9}y^{15}+121428246528000x^{9}y^{14}+269401404211200x^{9}y^{13}\\
&+266414063616000x^{9}y^{12}+76034919628800x^{9}y^{11}-136696194662400x^{9}y^{10}\\
&-372015208857600x^{9}y^{9}-370990330675200x^{9}y^{8}-612293699174400x^{9}y^{7}\\
&-661342283366400x^{9}y^{6}-376286714265600x^{9}y^{5}-220639002624000x^{9}y^{4}\\
&+78466881945600x^{9}y^{3}+135785034547200x^{9}y^{2}+51279298560000x^{9}y\\
&+135970947072000x^{8}y^{16}+62302401331200x^{8}y^{15}+66343403520000x^{8}y^{14}\\
&-136599463526400x^{8}y^{13}-437845229568000x^{8}y^{12}-652937527296000x^{8}y^{11}\\
&-629002469376000x^{8}y^{10}-471636954316800x^{8}y^{9}-284385293107200x^{8}y^{8}\\
&+79565134233600x^{8}y^{7}+165091501670400x^{8}y^{6}+88142354841600x^{8}y^{5}\\
&+62300985753600x^{8}y^{4}+14804110540800x^{8}y^{3}+80055160012800x^{8}y^{2}\\
&+52515097804800x^{8}y+4843162828800x^{7}y^{17}-362387865600x^{7}y^{16}\\
&-106537313894400x^{7}y^{15}-408752750592000x^{7}y^{14}-753768176025600x^{7}y^{13}\\
&-920964877516800x^{7}y^{12}-762648566169600x^{7}y^{11}-594930460262400x^{7}y^{10}\\
&-149366557900800x^{7}y^{9}+183090570854400x^{7}y^{8}+274517301657600x^{7}y^{7}\\
&+463521683865600x^{7}y^{6}+207495364608000x^{7}y^{5}+21700804608000x^{7}y^{4}\\
&+17296942694400x^{7}y^{3}+13169118412800x^{7}y^{2}+24639543705600x^{7}y\\
&-1166435942400x^{6}y^{18}-32686630502400x^{6}y^{17}-172273906483200x^{6}y^{16}\\
&-453954974515200x^{6}y^{15}-730544681779200x^{6}y^{14}-781666851225600x^{6}y^{13}\\
&-585238472294400x^{6}y^{12}-283342956134400x^{6}y^{11}+162148751769600x^{6}y^{10}\\
&+287331110092800x^{6}y^{9}+400885088256000x^{6}y^{8}+342099807436800x^{6}y^{7}\\
&+99965140992000x^{6}y^{6}+90285539328000x^{6}y^{5}-67067235532800x^{6}y^{4}\\
&-105251025715200x^{6}y^{3}-48022054502400x^{6}y^{2}-4127824281600x^{6}y\\
&-3193543065600x^{5}y^{19}-34256977920000x^{5}y^{18}-141998476492800x^{5}y^{17}\\
&-310779681177600x^{5}y^{16}-408066903244800x^{5}y^{15}-282698396467200x^{5}y^{14}\\
&+158698045440000x^{5}y^{13}+418131895910400x^{5}y^{12}+702850557542400x^{5}y^{11}\\
&+890508843417600x^{5}y^{10}+1029447784857600x^{5}y^{9}+684715121049600x^{5}y^{8}\\
&+492144309043200x^{5}y^{7}+277100022988800x^{5}y^{6}-19972030464000x^{5}y^{5}\\
&-136263499776000x^{5}y^{4}-105817964544000x^{5}y^{3}-93368667340800x^{5}y^{2}\\
&-25537727692800x^{5}y-2615987404800x^{4}y^{20}-21494130278400x^{4}y^{19}\\
&-72046293811200x^{4}y^{18}-107745273446400x^{4}y^{17}-7322782924800x^{4}y^{16}\\
&+309078864691200x^{4}y^{15}+766314676224000x^{4}y^{14}+1155354800947200x^{4}y^{13}\\
&+1198288561766400x^{4}y^{12}+1049523624345600x^{4}y^{11}+526319655321600x^{4}y^{10}\\
&+66560104857600x^{4}y^{9}-43546823884800x^{4}y^{8}-276861970022400x^{4}y^{7}\\
&-197625367756800x^{4}y^{6}-100720469606400x^{4}y^{5}-109583400960000x^{4}y^{4}\\
&-41772987187200x^{4}y^{3}-52517221171200x^{4}y^{2}-26257548902400x^{4}y\\
&-1343855001600x^{3}y^{21}-8499127910400x^{3}y^{20}-13181858611200x^{3}y^{19}\\
&+45014895820800x^{3}y^{18}+268101432115200x^{3}y^{17}+681171930316800x^{3}y^{16}\\
&+1153234737561600x^{3}y^{15}+1407795226214400x^{3}y^{14}+1324284641280000x^{3}y^{13}\\
&+916705404518400x^{3}y^{12}+218947033497600x^{3}y^{11}-136127014502400x^{3}y^{10}\\
&-537657547161600x^{3}y^{9}-616491122688000x^{3}y^{8}-302016724992000x^{3}y^{7}\\
&-286886028902400x^{3}y^{6}-78448479436800x^{3}y^{5}+24546823372800x^{3}y^{4}\\
&+3110731776000x^{3}y^{3}-4520647065600x^{3}y^{2}-12319771852800x^{3}y\\
&-460534579200x^{2}y^{22}-1096600780800x^{2}y^{21}+14318095564800x^{2}y^{20}\\
&+102259438387200x^{2}y^{19}+332003436134400x^{2}y^{18}+681460472217600x^{2}y^{17}\\
&+984398443315200x^{2}y^{16}+1031470527283200x^{2}y^{15}+771840265420800x^{2}y^{14}\\
&+216461397196800x^{2}y^{13}-287464292352000x^{2}y^{12}-640148845363200x^{2}y^{11}\\
&-964338765004800x^{2}y^{10}-710955034214400x^{2}y^{9}-552832539033600x^{2}y^{8}\\
&-331873085030400x^{2}y^{7}+42029088768000x^{2}y^{6}+37763363635200x^{2}y^{5}\\
&+112506214809600x^{2}y^{4}+101373758668800x^{2}y^{3}+36779891097600x^{2}y^{2}\\
&+2063912140800x^{2}y-75497472000xy^{23}+1317430886400xy^{22}+17813628518400xy^{21}\\
&+86621552640000xy^{20}+232081116364800xy^{19}+384929641267200xy^{18}+384578342092800xy^{17}\\
&+113521773772800xy^{16}-420487770931200xy^{15}-1019400722841600xy^{14}\\
&-1408288731955200xy^{13}-1688657146675200xy^{12}-1464426351820800xy^{11}\\
&-1053801204940800xy^{10}-863990651289600xy^{9}-325525929984000xy^{8}-128474696908800xy^{7}\\
&-28188278784000xy^{6}+164348382412800xy^{5}+87152866099200xy^{4}+91936456704000xy^{3}\\
&+59813108121600xy^{2}+12768863846400xy-7549747200y^{24}-35861299200y^{23}+796498329600y^{22}\\
&+9025250918400y^{21}+38865154867200y^{20}+90213580800000y^{19}+118775336140800y^{18}\\
&+58420651622400y^{17}-110741697331200y^{16}-349766162841600y^{15}-575277731020800y^{14}\\
&-657489936384000y^{13}-716383678464000y^{12}-587997090201600y^{11}-313230845952000y^{10}\\
&-297224920166400y^{9}-81977504716800y^{8}+9969848985600y^{7}-41270811379200y^{6}\\
&+70192637164800y^{5}+21071447827200y^{4}+25579068579072y^{3}+20093063539392y^{2}\\
&+5747258854944y+637578926856,\\
&1834588569600x^{22}y^{2}+1834588569600x^{22}y+5503765708800x^{21}y^{3}\\
&+12536355225600x^{21}y^{2}+7032589516800x^{21}y+2751882854400x^{20}y^{4}
\end{aligned}
$$

$$+40055183769600x^{20}y^3 + 27518828544000x^{20}y^2 - 9784472371200x^{20}y$$
$$-6319138406400x^{19}y^5 + 46476243763200x^{19}y^4 + 58299147878400x^{19}y^3$$
$$-28334201241600x^{19}y^2 - 33837966950400x^{19}y - 5503765708800x^{18}y^6$$
$$-1019215872000x^{18}y^5 + 5809530470400x^{18}y^4 - 76594072780800x^{18}y^3$$
$$-103501371801600x^{18}y^2 - 25582318387200x^{18}y + 7813988352000x^{17}y^7$$
$$-70291921305600x^{17}y^6 - 179517888921600x^{17}y^5 - 185174537011200x^{17}y^4$$
$$-162887683276800x^{17}y^3 - 39324745728000x^{17}y^2 + 39800379801600x^{17}y$$
$$+11754956390400x^{16}y^8 - 120488303001600x^{16}y^7 - 317655613440000x^{16}y^6$$
$$-385110717235200x^{16}y^5 - 128811899289600x^{16}y^4 + 109344876134400x^{16}y^3$$
$$+137322351820800x^{16}y^2 + 98863939584000x^{16}y - 7293055795200x^{15}y^9$$
$$-114582513254400x^{15}y^8 - 287316954316800x^{15}y^7 - 301200939417600x^{15}y^6$$
$$-46946215526400x^{15}y^5 + 252139850956800x^{15}y^4 + 284267800166400x^{15}y^3$$
$$+154258322227200x^{15}y^2 + 47903145984000x^{15}y - 27983137996800x^{14}y^{10}$$
$$-53542807142400x^{14}y^9 - 66837911961600x^{14}y^8 + 149224528281600x^{14}y^7$$
$$+151696126771200x^{14}y^6 + 298576458547200x^{14}y^5 + 302067272908800x^{14}y^4$$
$$-9257877504000x^{14}y^3 - 32495999385600x^{14}y^2 - 58553951846400x^{14}y$$
$$-23657132851200x^{13}y^{11} + 29221296537600x^{13}y^{10} + 287835999436800x^{13}y^9$$
$$+573753419366400x^{13}y^8 + 520178525798400x^{13}y^7 + 120040036761600x^{13}y^6$$
$$-111556008345600x^{13}y^5 - 201634873344000x^{13}y^4 - 248705659699200x^{13}y^3$$
$$-214417539072000x^{13}y^2 - 117133384089600x^{13}y + 98146713600x^{12}y^{12}$$
$$+110758566297600x^{12}y^{11} + 511160352768000x^{12}y^{10} + 765609482649600x^{12}y^9$$
$$+685166926233600x^{12}y^8 - 9043653427200x^{12}y^7 - 401507824435200x^{12}y^6$$
$$-221825256652800x^{12}y^5 - 253633285324800x^{12}y^4 - 131801599180800x^{12}y^3$$
$$-92306984140800x^{12}y^2 - 64720207872000x^{12}y + 20414516428800x^{11}y^{13}$$
$$+152229327667200x^{11}y^{12} + 448762635878400x^{11}y^{11} + 630349155532800x^{11}y^{10}$$
$$+406131100876800x^{11}y^9 - 145656329011200x^{11}y^8 - 338977043251200x^{11}y^7$$
$$-177438405427200x^{11}y^6 + 88392912076800x^{11}y^5 + 235604489011200x^{11}y^4$$
$$+125962341580800x^{11}y^3 + 83626662297600x^{11}y^2 + 28028436480000x^{11}y$$
$$+24468730675200x^{10}y^{14} + 118170530611200x^{10}y^{13} + 202178455142400x^{10}y^{12}$$
$$+177636114432000x^{10}y^{11} - 107596165939200x^{10}y^{10} - 302728819507200x^{10}y^9$$
$$-313834969497600x^{10}y^8 - 347999934873600x^{10}y^7 - 206737558732800x^{10}y^6$$
$$-230315891097600x^{10}y^5 - 126360118886400x^{10}y^4 + 131330211840000x^{10}y^3$$
$$+141981017702400x^{10}y^2 + 68007179059200x^{10}y + 14729556787200x^9y^{15}$$
$$+35410201804800x^9y^{14} - 65769622732800x^9y^{13} - 327852490752000x^9y^{12}$$
$$-599199842304000x^9y^{11} - 565304780390400x^9y^{10} - 443513202278400x^9y^9$$
$$-146677432320000x^9y^8 - 12724627046400x^9y^7 - 65767027507200x^9y^6$$
$$-20913035673600x^9y^5 - 109913938329600x^9y^4 - 21756012134400x^9y^3$$
$$+90646511616000x^9y^2 + 59687829504000x^9y + 1570347417600x^8y^{16}$$
$$-36820117094400x^8y^{15} - 234266768179200x^8y^{14} - 581083752038400x^8y^{13}$$
$$-805240936857600x^8y^{12} - 743508541440000x^8y^{11} - 551910585139200x^8y^{10}$$
$$-81281522073600x^8y^9 + 238031968665600x^8y^8 + 640740202905600x^8y^7$$
$$+784672594329600x^8y^6 + 374951116800000x^8y^5 + 154749527654400x^8y^4$$
$$-27060181401600x^8y^3 - 23577860505600x^8y^2 + 18091081728000x^8y$$
$$-6254965555200x^7y^{17} - 66554796441600x^7y^{16} - 256693292236800x^7y^{15}$$
$$-523739175321600x^7y^{14} - 629637591859200x^7y^{13} - 523208805580800x^7y^{12}$$
$$-247251389644800x^7y^{11} + 1865259417600x^7y^{10} + 191297853849600x^7y^9$$
$$+343814071910400x^7y^8 + 203426050867200x^7y^7 + 192544033996800x^7y^6$$
$$+121463635968000x^7y^5 - 64839116390400x^7y^4 - 65769150873600x^7y^3$$
$$-66779873280000x^7y^2 - 17479552204800x^7y - 7225108070400x^6y^{18}$$
$$-56709926092800x^6y^{17} - 172690086297600x^6y^{16} - 273202701926400x^6y^{15}$$
$$-194329549209600x^6y^{14} + 56835440640000x^6y^{13} + 350209651507200x^6y^{12}$$
$$+444519206092800x^6y^{11} + 599186630246400x^6y^{10} + 490139497267200x^6y^9$$
$$+272471084236800x^6y^8 + 225555657523200x^6y^7 - 50499905126400x^6y^6$$
$$-59515836825600x^6y^5 - 34242822144000x^6y^4 - 124539686092800x^6y^3$$
$$-92986461388800x^6y^2 - 33188216832000x^6y - 4745016115200x^5y^{19}$$
$$-30274486272000x^5y^{18} - 64811748556800x^5y^{17} - 9451339776000x^5y^{16}$$
$$+243667619020800x^5y^{15} + 653881953484800x^5y^{14} + 982164307968000x^5y^{13}$$
$$+1068378410188800x^5y^{12} + 942097563648000x^5y^{11} + 485977227264000x^5y^{10}$$
$$+226324670054400x^5y^9 - 87462995558400x^5y^8 - 242420613120000x^5y^7$$
$$-101214152294400x^5y^6 - 149051473920000x^5y^5 - 37652948582400x^5y^4$$
$$-16655686041600x^5y^3 - 55354038681600x^5y^2 - 29741993164800x^5y$$
$$-2004457881600x^4y^{20} - 7832862720000x^4y^{19} + 10945246003200x^4y^{18}$$
$$+145556294860800x^4y^{17} + 461781467136000x^4y^{16} + 886536850636800x^4y^{15}$$
$$+1198437433344000x^4y^{14} + 1241889059635200x^4y^{13} + 854355581337600x^4y^{12}$$
$$+256056636211200x^4y^{11} - 318164041728000x^4y^{10} - 872748534988800x^4y^9$$
$$-779903631360000x^4y^8 - 635737197772800x^4y^7 - 471803048755200x^4y^6$$
$$-104551022592000x^4y^5 - 36412902604800x^4y^4 + 37500773990400x^4y^3$$
$$+20392810905600x^4y^2 - 9045540864000x^4y - 392586854400x^3y^{21}$$
$$+2806618521600x^3y^{20} + 38964245299200x^3y^{19} + 170372313907200x^3y^{18}$$
$$+418705676697600x^3y^{17} + 685531201536000x^3y^{16} + 802542374092800x^3y^{15}$$
$$+637971569049600x^3y^{14} + 233189513625600x^3y^{13} - 107876686233600x^3y^{12}$$
$$-481140198604800x^3y^{11} - 550602827366400x^3y^{10} - 462367398297600x^3y^9$$

$$
\begin{aligned}
& -480060289843200x^3y^8 - 133128408268800x^3y^7 - 8291627827200x^3y^6 \\
& +24260876697600x^3y^5 + 137759057510400x^3y^4 + 88219503820800x^3y^3 \\
& +49984045056000x^3y^2 + 8739776102400x^3y + 150994944000x^2y^{22} \\
& +4627995033600x^2y^{21} + 33339683635200x^2y^{20} + 113284428595200x^2y^{19} \\
& +218615198515200x^2y^{18} + 233608996454400x^2y^{17} + 35591159808000x^2y^{16} \\
& -386561566310400x^2y^{15} - 806588920627200x^2y^{14} - 1088800948224000x^2y^{13} \\
& -1167544575590400x^2y^{12} - 818457123225600x^2y^{11} - 633404620800000x^2y^{10} \\
& -266152712601600x^2y^9 + 47717351424000x^2y^8 - 16605079142400x^2y^7 \\
& +176811422515200x^2y^6 + 156384873676800x^2y^5 + 88412022374400x^2y^4 \\
& +106540970803200x^2y^3 + 61364227276800x^2y^2 + 16594108416000x^2y + 173644185600xy^{23} \\
& +2957613465600xy^{22} + 16862360371200xy^{21} + 40334996275200xy^{20} + 12344780390400xy^{19} \\
& -192792939724800xy^{18} - 655032110284800xy^{17} - 1271857899110400xy^{16} \\
& -1777243874918400xy^{15} - 1978056533606400xy^{14} - 1644003223142400xy^{13} \\
& -1029274081689600xy^{12} - 387784684339200xy^{11} + 360973644595200xy^{10} + 454042681344000xy^9 \\
& +491194810368000xy^8 + 453087579340800xy^7 + 130788104601600xy^6 + 128771024486400xy^5 \\
& +42624103219200xy^4 + 13002434150400xy^3 + 3219978977280 0xy^2 + 14870996582400xy \\
& +103809024000y^{23} + 1475975577600y^{22} + 7233601536000y^{21} + 12385360281600y^{20} \\
& -16787452723200y^{19} - 131230413619200y^{18} - 343940412211200y^{17} - 586190094336000y^{16} \\
& -737172180172800y^{15} - 762550154035200y^{14} - 571910941900800y^{13} - 263618872934400y^{12} \\
& -43160109465600y^{11} + 297717931008000y^{10} + 290972696371200y^9 + 217489096857600y^8 \\
& +240414438067200y^7 + 47308019328000y^6 + 27864316051200y^5 + 22561237411200y^4 \\
& -15252543046656y^3 + 11686347969984y^2 + 1776055783248y + 1737327290832\rangle \\
B \quad & \langle 22649241600x^3y^{24} + 1253258035200x^3y^{23} + 10311067238400x^3y^{22} \\
& +34490076364800x^3y^{21} + 42036992409600x^3y^{20} - 67678057267200x^3y^{19} \\
& -390062171750400x^3y^{18} - 905004358041600x^3y^{17} - 1450137511526400x^3y^{16} \\
& -1711475018956800x^3y^{15} - 1610665810329600x^3y^{14} - 1177861924454400x^3y^{13} \\
& -299175697612800x^3y^{12} + 112906775347200x^3y^{11} + 475672828723200x^3y^{10} \\
& +807384341606400x^3y^9 + 403248945715200x^3y^8 + 377108933990400x^3y^7 \\
& +255035002406400x^3y^6 + 15039290131200x^3y^5 + 71552271801600x^3y^4 \\
& +10696522998528x^3y^3 + 12325432972608x^3y^2 + 7381515596256x^3y - 637578926856x^3 \\
& +15099494400x^2y^{25} + 588880281600x^2y^{24} + 3554987212800x^2y^{23} \\
& +2741501952000x^2y^{22} - 46700140953600x^2y^{21} - 233784528076800x^2y^{20} \\
& -610208081510400x^2y^{19} - 1096723228262400x^2y^{18} - 1466410283827200x^2y^{17} \\
& -1506767339520000x^2y^{16} - 1232157052108800x^2y^{15} - 577335759667200x^2y^{14} \\
& +26683534540800x^2y^{13} + 347152435200000x^2y^{12} + 893048796979200x^2y^{11} \\
& +644131836518400x^2y^{10} + 435211525785600x^2y^9 + 454968193766400x^2y^8 \\
& -31632470092800x^2y^7 - 16784946144000x^2y^6 - 23843555107200x^2y^5 \\
& -90316217462400x^2y^4 - 19133687009664x^2y^3 - 4934349023904x^2y^2 \\
& +2469128128272x^2y + 318789463428x^2 + 3774873600xy^{26} + 173644185600xy^{25} \\
& +110415052800xy^{24} - 102775652352 00xy^{23} - 70653837312000xy^{22} - 239790234009600xy^{21} \\
& -517202981683200xy^{20} - 776757628108800xy^{19} - 823677065625600xy^{18} - 571067655782400xy^{17} \\
& -267616124928 00xy^{16} + 642389277081600xy^{15} + 1068460181913600xy^{14} \\
& +1523999571148800xy^{13} + 1539853684531200xy^{12} + 1032314214297600xy^{11} \\
& +1034421402931200xy^{10} + 445571816832000xy^9 + 73284619507200xy^8 + 120712127731200xy^7 \\
& -201736249257600xy^6 - 119613162196800xy^5 - 76485128644800xy^4 - 72058545960768xy^3 \\
& -22307202960048xy^2 - 2266520134536xy - 159394731714x + 13212057600y^{26} - 828584755200y^{25} \\
& -9934995456000y^{24} - 48043288166400y^{23} - 130147024896000y^{22} - 218361456230400y^{21} \\
& -216453729484800y^{20} - 383002214400 00y^{19} + 341339937177600y^{18} + 866275953868800y^{17} \\
& +1343237573836800y^{16} + 1723451034009600y^{15} + 1879537703731200y^{14} + 1558649924812800y^{13} \\
& +1326809245593600y^{12} + 967191965337600y^{11} + 385238501683200y^{10} + 328525882560000y^9 \\
& +53055678931200y^8 - 101233933977600y^7 - 9678715963200y^6 - 108778116434400y^5 \\
& -54414389000736y^4 - 26369919064512y^3 - 21626582008248y^2 - 5569961319360y + 79697365857, \\
& 83047219200x^3y^{24} + 1007891251200x^3y^{23} + 2268699033600x^3y^{22} \\
& -16078130380800x^3y^{21} - 118647580262400x^3y^{20} - 376976808345600x^3y^{19} \\
& -766247672217600x^3y^{18} - 1128680364441600x^3y^{17} - 1286508065587200x^3y^{16} \\
& -1088832326860800x^3y^{15} - 518087806156800x^3y^{14} + 25645503283200x^3y^{13} \\
& +662977437696000x^3y^{12} + 1010209882521600x^3y^{11} + 794286882201600x^3y^{10} \\
& +934385100595200x^3y^9 + 525324071577600x^3y^8 + 151821471052800x^3y^7 \\
& +175915356134400x^3y^6 - 73415125958400x^3y^5 - 54138173040000x^3y^4 \\
& -23967096044544x^3y^3 - 26252641473984x^3y^2 + 2746714648752x^3y - 1737327290832x^3 \\
& +26424115200x^2y^{25} + 132120576000x^2y^{24} - 2454611558400x^2y^{23} \\
& -26447708160000x^2y^{22} - 116073588326400x^2y^{21} - 296106983424000x^2y^{20} \\
& -496153106841600x^2y^{19} - 569060425728000x^2y^{18} - 389893953945600x^2y^{17} \\
& +539872395264 00x^2y^{16} + 481552072704000x^2y^{15} + 776596414464000x^2y^{14} \\
& +968889987072000x^2y^{13} + 638895624192000x^2y^{12} + 524343905280000x^2y^{11} \\
& +318017421619200x^2y^{10} - 189131084697600x^2y^9 - 22272452121600x^2y^8 \\
& -178758434880000x^2y^7 - 274993581657600x^2y^6 - 88730660995200x^2y^5 \\
& -124442838216000x^2y^4 - 67178205577728x^2y^3 - 20162755999008x^2y^2 \\
& -5743245375576x^2y + 868663645416x^2 + 3774873600xy^{26} - 115133644800xy^{25} \\
& -2776419532800xy^{24} - 19786471833600xy^{23} - 69364953907200xy^{22} - 131447047808 00xy^{21} \\
& -102810923827200xy^{20} - 150150433996800xy^{19} + 674618867712000xy^{18} \\
& +1306894447411200xy^{17} + 1787499882086400xy^{16} + 2065757257728000xy^{15}
\end{aligned}
$$

$+1826925966950400xy^{14} + 1298669930496000xy^{13} + 906874347110400xy^{12}$
$+75130639257600xy^{11} - 85046196787200xy^{10} - 98017154611200xy^{9} - 405845513049600xy^{8}$
$-90110743315200xy^{7} - 112770929260800xy^{6} - 139091756990400xy^{5} - 27464977000800xy^{4}$
$-44078331707136xy^{3} - 28036233930096xy^{2} - 5425431520212xy - 434331822708x - 96259276800y^{26}$
$-1549585612800y^{25} - 8641629388800y^{24} - 19601974886400y^{23} + 3114860544000y^{22}$
$+141056055705600y^{21} + 463132753920000y^{20} + 947310546124800y^{19} + 1455000374476800y^{18}$
$+1844055333273600y^{17} + 1920998478643200y^{16} + 1572973785907200y^{15} + 1069517880115200y^{14}$
$+317473916313600y^{13} - 325003806720000y^{12} - 465271484774400y^{11} - 755613916185600y^{10}$
$-578695153766400y^{9} - 292661432236800y^{8} - 290839991529600y^{7} - 50754014380800y^{6}$
$-28721364415200y^{5} - 23488684763472y^{4} + 2259276502176y^{3} - 2969805812976y^{2}$
$-5591446176510y + 217165911354\rangle$

$i = 11,\ D_i = 16,\ \rho_{ij}$: 15:2, 20:2, 22:2, 24:2, 30:2, 31:1, 34:1, 42:1, 43:2, 44:2

W $\langle 84y^2 - 18y + 72,$
$144y^2 + 6y + 45\rangle$

A $\langle 11943936x^{13} + 13934592x^{12}y + 31850496x^{12} + 13934592x^{11}y^2 - 9953280x^{11}y$
$+49766400x^{11} - 36495360x^{10}y^3 - 54411264x^{10}y^2 - 95551488x^{10}y + 29859840x^{10}$
$-54411264x^{9}y^4 - 199729152x^{9}y^3 - 244518912x^{9}y^2 - 232906752x^{9}y - 19906560x^{9}$
$-64806912x^{8}y^5 - 119549952x^{8}y^4 - 338411520x^{8}y^3 - 397135872x^{8}y^2$
$-328458240x^{8}y - 71663616x^{8} + 8626176x^{7}y^6 - 1437696x^{7}y^5 + 24440832x^{7}y^4$
$-69672960x^{7}y^3 - 325804032x^{7}y^2 - 278691840x^{7}y - 78630912x^{7} + 48660480x^{6}y^7$
$+246730752x^{6}y^6 + 426332160x^{6}y^5 + 492687360x^{6}y^4 + 513091584x^{6}y^3$
$+69672960x^{6}y^2 - 107495424x^{6}y - 40808448x^{6} + 78962688x^{5}y^8 + 255246336x^{5}y^7$
$+591888384x^{5}y^6 + 834416640x^{5}y^5 + 821090304x^{5}y^4 + 860792832x^{5}y^3$
$+445741056x^{5}y^2 + 92565504x^{5}y + 13934592x^{5} + 41803776x^{4}y^9 + 176394240x^{4}y^8$
$+335591424x^{4}y^7 + 487157760x^{4}y^6 + 665542656x^{4}y^5 + 542121984x^{4}y^4$
$+728580096x^{4}y^3 + 523708416x^{4}y^2 + 189609984x^{4}y + 47775744x^{4} + 9289728x^{3}y^{10}$
$-9621504x^{3}y^9 - 50485248x^{3}y^8 - 131328000x^{3}y^7 - 144211968x^{3}y^6$
$+65581056x^{3}y^5 - 40172544x^{3}y^4 + 134866944x^{3}y^3 + 298432512x^{3}y^2$
$+151787520x^{3}y + 47278080x^{3} - 19243008x^{2}y^{11} - 102629376x^{2}y^{10} - 323426304x^{2}y^9$
$-618762240x^{2}y^8 - 903204864x^{2}y^7 - 1093008384x^{2}y^6 - 842545152x^{2}y^5$
$-660704256x^{2}y^4 - 458348544x^{2}y^3 - 64696320x^{2}y^2 + 36827136x^{2}y + 22394880x^{2}$
$-23224320xy^{12} - 126406656xy^{11} - 342392832xy^{10} - 683624448xy^{9} - 1061102592xy^{8}$
$-1312422912xy^{7} - 1560563712xy^{6} - 1330919424xy^{5} - 1013022720xy^{4} - 750809088xy^{3}$
$-323315712xy^{2} - 74151936xy - 6967296x - 442368y^{13} + 9842688y^{12} + 64143360y^{11}$
$+189139968y^{10} + 410628096y^{9} + 667395072y^{8} + 861262848y^{7} + 1012013568y^{6} + 795121920y^{5}$
$+614525184y^{4} + 395065728y^{3} + 100707840y^{2} - 1609632y - 9811368,$
$-13934592x^{12}y - 1990656x^{12} - 33841152x^{11}y^2 - 69672960x^{11}y - 17915904x^{11}$
$-59056128x^{10}y^3 - 114794496x^{10}y^2 - 175177728x^{10}y - 45785088x^{10} - 30523392x^{9}y^4$
$-138018816x^{9}y^3 - 150294528x^{9}y^2 - 215986176x^{9}y - 65691648x^{9} + 19021824x^{8}y^5$
$+50540544x^{8}y^4 - 30191616x^{8}y^3 - 16588800x^{8}y^2 - 113467392x^{8}y - 44789760x^{8}$
$+92454912x^{7}y^6 + 192983040x^{7}y^5 + 398462976x^{7}y^4 + 294948864x^{7}y^3$
$+224944128x^{7}y^2 + 101523456x^{7}y + 16920576x^{7} + 90021888x^{6}y^7 + 293621760x^{6}y^6$
$+434405376x^{6}y^5 + 628273152x^{6}y^4 + 584755200x^{6}y^3 + 340899840x^{6}y^2$
$+251817984x^{6}y + 76640256x^{6} + 60825600x^{5}y^8 + 138461184x^{5}y^7 + 254361600x^{5}y^6$
$+232796160x^{5}y^5 + 271890432x^{5}y^4 + 412563456x^{5}y^3 + 213829632x^{5}y^2$
$+223948800x^{5}y + 94556160x^{5} - 7962624x^{4}y^9 - 14155776x^{4}y^8 - 167712768x^{4}y^7$
$-227874816x^{4}y^6 - 371589120x^{4}y^5 - 376897536x^{4}y^4 - 62042112x^{4}y^3$
$-83773440x^{4}y^2 + 58226688x^{4}y + 60217344x^{4} - 32514048x^{3}y^{10} - 150405120x^{3}y^9$
$-375072768x^{3}y^8 - 662390784x^{3}y^7 - 763416576x^{3}y^6 - 780503040x^{3}y^5$
$-881556480x^{3}y^4 - 478006272x^{3}y^3 - 323149824x^{3}y^2 - 114960384x^{3}y + 5474304x^{3}$
$-43130880x^{2}y^{11} - 142553088x^{2}y^{10} - 350300160x^{2}y^9 - 565512192x^{2}y^8$
$-856590336x^{2}y^7 - 987393024x^{2}y^6 - 831237120x^{2}y^5 - 871188480x^{2}y^4$
$-580773888x^{2}y^3 - 343305216x^{2}y^2 - 193093632x^{2}y - 36329472x^{2} - 25214976xy^{12}$
$-90132480xy^{11} - 164450304xy^{10} - 234233856xy^{9} - 185711616xy^{8} - 223782912xy^{7}$
$-332494848xy^{6} - 253974528xy^{5} - 343498752xy^{4} - 319002624xy^{3} - 176007168xy^{2} - 146064384xy$
$-47278080x + 442368y^{14} + 1548288y^{13} + 15814656y^{12} + 62760960y^{11} + 132793344y^{10}$
$+236141568y^{9} + 294838272y^{8} + 425392128y^{7} + 513644544y^{6} + 444642048y^{5} + 497729664y^{4}$
$+344518272y^{3} + 204550272y^{2} + 120772944y + 24898752\rangle$

B $\langle -16146432x^{3}y^{13} - 89800704x^{3}y^{12} - 276092928x^{3}y^{11} - 549586944x^{3}y^{10}$
$-913876992x^{3}y^9 - 1318920192x^{3}y^8 - 1586442240x^{3}y^7 - 1965537792x^{3}y^6$
$-1754742528x^{3}y^5 - 1308351744x^{3}y^4 - 1027928448x^{3}y^3 - 459025920x^{3}y^2$
$-116834400x^{3}y - 14076504x^{3} - 8847360x^{2}y^{14} - 40144896x^{2}y^{13} - 73156608x^{2}y^{12}$
$-91183104x^{2}y^{11} - 59553792x^{2}y^{10} - 37269504x^{2}y^9 - 153916416x^{2}y^8$
$-180928512x^{2}y^7 - 529659648x^{2}y^6 - 797344128x^{2}y^5 - 631794816x^{2}y^4$
$-651236544x^{2}y^3 - 420781824x^{2}y^2 - 145508400x^{2}y - 30677292x^{2} - 2654208xy^{15}$
$-2985984xy^{14} + 40531968xy^{13} + 227045376xy^{12} + 596146176xy^{11} + 1120573440xy^{10}$
$+1669745664xy^9 + 1853314560xy^8 + 1894814208xy^7 + 1510641792xy^6 + 793509696xy^5$
$+405342144xy^4 + 35602848xy^3 - 152164224xy^2 - 87684120xy - 26536086x - 884736y^{16}$
$+8073216y^{15} + 72216576y^{14} + 250629120y^{13} + 606514176y^{12} + 1138627584y^{11}$
$+1714701312y^{10} + 2281692672y^9 + 2472761088y^8 + 2269582848y^7 + 2019515328y^6$
$+1291572000y^5 + 710383392y^4 + 383411664y^3 + 89864208y^2 - 1860408y - 9784395,$
$-442368x^{3}y^{14} - 14819328x^{3}y^{13} - 33730560x^{3}y^{12} - 29693952x^{3}y^{11}$

$+66963456x^3y^{10} + 199839744x^3y^9 + 409743360x^3y^8 + 468163584x^3y^7$
$+226824192x^3y^6 + 257935104x^3y^5 - 71217792x^3y^4 - 216452736x^3y^3$
$-131974272x^3y^2 - 152623440x^3y - 55007424x^3 - 3317760x^2y^{14} + 8404992x^2y^{13}$
$+116066304x^2y^{12} + 362078208x^2y^{11} + 797810688x^2y^{10} + 1222760448x^2y^9$
$+1642429440x^2y^8 + 1928931840x^2y^7 + 1611975168x^2y^6 + 1367114112x^2y^5$
$+914832576x^2y^4 + 354518208x^2y^3 + 189397440x^2y^2 - 666792x^2y - 30240864x^2$
$-221184xy^{15} + 23003136xy^{14} + 123144192xy^{13} + 371644416xy^{12} + 771268608xy^{11}$
$+1331527680xy^{10} + 1827698688xy^9 + 2141766144xy^8 + 2419511040xy^7 + 2156633856xy^6$
$+1736061120xy^5 + 1322259552xy^4 + 670361184xy^3 + 357548256xy^2 + 119852460xy + 3044304x$
$+221184y^{16} + 14266368y^{15} + 75036672y^{14} + 191794176y^{13} + 378114048y^{12} + 587105280y^{11}$
$+847715328y^{10} + 1171196928y^9 + 1250028288y^8 + 1365994368y^7 + 1353272832y^6$
$+993215520y^5 + 821655792y^4 + 489078000y^3 + 230030280y^2 + 120509046y + 25161192\rangle$

$i = 12,\ D_i = 16,\ \rho_{ij}$: 6:2, 8:1, 10:1, 17:2, 23:2, 29:2, 33:2, 34:1, 44:1, 49:2, 54:2
$w_i \quad \langle -11254t^3 + 3862t^2 - 788t - 766,$
$-3098t^3 + 914t^2 - 76t - 2150\rangle$

$i = 13,\ D_i = 2,\ \rho_{ij}$: 21:2, 46:2, 51:1, 52:1
$w_i \quad \langle -20t^3 - 20t^2 + 30t + 10,$
$-10t^3 - 40t^2 + 20\rangle$

$i = 14,\ D_i = 64,\ \rho_{ij}$: 3:2, 5:2, 7:2, 10:1, 16:2, 22:2, 24:2, 26:1, 27:2, 29:2, 43:2, 47:2, 51:1, 53:1, 54:2
$w_i \quad \langle 73728t^3 + 91296t^2 - 99990t - 11790,$
$83538t^3 - 80100t^2 - 72252t + 22320\rangle$

$i = 15,\ D_i = 2,\ \rho_{ij}$: 8:1, 11:1, 28:2, 42:2, 58:1
$w_i \quad \langle -22t^2 - 2t,$
$4t^2 + 8t\rangle$

$i = 16,\ D_i = 8,\ \rho_{ij}$: 6:2, 7:1, 14:1, 24:1, 25:2, 30:1, 34:1, 38:2, 40:2, 43:1, 51:2, 53:1
$w_i \quad \langle -1554t^3 - 2148t^2 + 2172t - 714,$
$-3018t^3 + 1806t^2 + 804t - 444\rangle$

$i = 17,\ D_i = 16,\ \rho_{ij}$: 3:2, 5:1, 8:1, 9:1, 10:1, 12:1, 22:2, 27:1, 28:1, 32:2, 34:2, 36:2, 39:1, 41:2, 47:2, 50:2, 51:2
$w_i \quad \langle -287462t^3 + 641736t^2 + 940208t - 79176,$
$765632t^3 + 1635156t^2 + 632158t - 440010\rangle$

$i = 18,\ D_i = 16,\ \rho_{ij}$: 3:2, 4:1, 6:2, 27:1, 34:1, 41:2, 49:1, 52:1, 54:1, 56:2
$w_i \quad \langle 296t^3 - 208t^2 - 106t + 78,$
$34t^3 + 172t^2 - 206t + 54\rangle$

$i = 19,\ D_i = 512,\ e_i = 0,\ \rho_{ij}$: 3:2, 4:2, 8:1, 28:2, 36:2, 50:1, 52:1
$w_i \quad \langle 16880t^3 + 58320t^2 + 12960t - 23440,$
$-8000t^3 + 44640t^2 + 59760t + 2080\rangle$

$i = 20,\ D_i = 8,\ \rho_{ij}$: 2:2, 10:1, 11:1, 22:2, 23:2, 27:2, 31:1, 33:1, 35:2, 36:1, 40:2, 52:1, 55:1, 57:1, 58:2
$w_i \quad \langle 1722t^3 + 1692t^2 - 324t + 978,$
$-1446t^3 + 2028t^2 - 996t + 450\rangle$

$i = 21,\ D_i = 8,\ \rho_{ij}$: 2:2, 13:1, 22:2, 25:1, 32:1, 34:2, 38:1, 43:2, 48:1, 52:2, 54:1, 58:1
$w_i \quad \langle 36t^3 + 48t^2 + 24t - 36,$
$30t^3 + 72t^2 + 36t + 18\rangle$

$i = 22,\ D_i = 16,\ \rho_{ij}$: 2:2, 3:2, 4:2, 5:2, 10:2, 11:1, 14:1, 17:1, 20:1, 21:1, 26:1, 40:1, 48:2, 50:2, 52:2
$w_i \quad \langle -3714t^3 - 827t^2 + 4772t - 67,$
$-381t^3 - 5045t^2 + 2208t + 996\rangle$

$i = 23,\ D_i = 8,\ \rho_{ij}$: 2:1, 8:1, 12:1, 20:1, 25:1, 26:1, 36:2, 49:1, 50:2, 51:1, 56:2
$w_i \quad \langle 12t^3 + 319t^2 + 246t - 23,$
$-279t^3 - 44t^2 + 283t + 113\rangle$

$i = 24,\ D_i = 8,\ \rho_{ij}$: 0:1, 1:1, 2:2, 10:1, 11:1, 14:1, 16:2, 28:2, 35:2, 38:2, 40:1, 42:2, 44:1, 48:2, 52:2
$w_i \quad \langle -533t^3 + 2798t^2 - 1063t + 403,$
$-2584t^3 + 1789t^2 + 439t - 250\rangle$

$i = 25,\ D_i = 16,\ \rho_{ij}$: 7:2, 16:1, 21:2, 23:2, 26:1, 27:2, 28:2, 30:1, 35:1, 38:1, 42:1, 44:1, 49:1, 57:2
$w_i \quad \langle 1518t^3 + 5544t^2 + 5817t + 1521,$
$-3303t^3 - 2058t^2 - 1008t + 366\rangle$

$i = 26,\ D_i = 8,\ \rho_{ij}$: 3:1, 5:1, 14:2, 22:2, 23:2, 25:2, 27:1, 46:1, 49:2, 50:1, 56:1
$w_i \quad \langle 1005t^3 + 4875t^2 + 4980t + 1965,$
$-360t^3 - 720t^2 - 480t - 120\rangle$

$i = 27,\ D_i = 8,\ \rho_{ij}$: 4:1, 5:2, 6:1, 14:1, 17:2, 18:2, 20:1, 25:1, 26:2, 28:2, 37:2, 41:1, 43:2, 52:1, 54:1, 55:2
$w_i \quad \langle -378t^3 + 1260t^2 - 126t - 90,$
$-1476t^3 + 1026t^2 + 324t - 162\rangle$

$i = 28,\ D_i = 64,\ \rho_{ij}$: 15:1, 17:2, 19:1, 24:1, 25:1, 27:1, 35:2, 37:1, 38:1, 40:1, 41:1, 46:1, 48:2, 49:1, 50:1, 51:2, 58:2
$w_i \quad \langle 8625t^3 - 7614t^2 - 6093t + 873,$
$3345t^3 - 9885t^2 + 1662t - 1281\rangle$

$i = 29,\ D_i = 32,\ \rho_{ij}$: 0:1, 5:1, 7:1, 8:2, 9:2, 10:1, 12:1, 14:1, 32:1, 33:2, 40:2, 45:1, 47:1, 50:2, 54:2
$w_i \quad \langle 718t^3 + 1083t^2 - 172t - 633,$
$167t^3 - 3t^2 - 965t - 414\rangle$

$i = 30,\ D_i = 16,\ \rho_{ij}$: 0:1, 7:1, 10:1, 11:1, 16:2, 25:2, 31:1, 44:1, 45:1, 46:2, 47:1, 48:2, 50:2, 51:1, 52:1
$w_i \quad \langle -1441t^3 - 7955t^2 + 4504t + 1622,$

$-6308t^3 - 55t^2 + 8108t - 281\rangle$

$i = 31,\ D_i = 2,\ \rho_{ij}$: 5:1, 9:1, 11:2, 20:2, 30:2, 40:2, 46:1, 47:1, 54:1, 58:1

$w_i \quad \langle 4t^3 - 48t^2 + 56t - 12,$
$68t^3 - 144t^2 + 112t - 36\rangle$

$i = 32,\ D_i = 2,\ \rho_{ij}$: 1:1, 5:1, 6:2, 8:1, 10:2, 17:1, 21:2, 29:2, 40:1, 42:1, 43:1

$w_i \quad \langle 16t^3 + 20t^2 + 2t + 10,$
$20t^3 + 22t^2 + 4t - 4\rangle$

$i = 33,\ D_i = 2,\ \rho_{ij}$: 2:2, 9:1, 10:1, 12:1, 20:2, 29:1, 43:2, 44:2, 49:2

$w_i \quad \langle -30t^3 + 84t^2 - 9t - 9,$
$-45t^3 + 66t^2 - 48t\rangle$

$i = 34,\ D_i = 16,\ \rho_{ij}$: 6:1, 11:2, 12:2, 16:2, 17:1, 18:2, 21:1, 35:2, 43:2, 46:2, 47:1, 48:2, 49:1, 54:1

$w_i \quad \langle 4200t^3 - 6405t^2 + 2994t + 591,$
$651t^3 - 4263t^2 + 4866t - 1440\rangle$

$i = 35,\ D_i = 16,\ \rho_{ij}$: 1:2, 10:2, 20:1, 24:1, 25:2, 28:1, 34:1, 44:2, 45:2, 49:1, 52:2, 53:2, 54:2

$w_i \quad \langle -2454t^3 + 339t^2 + 2742t - 879,$
$-1119t^3 + 3126t^2 + 291t - 399\rangle$

$i = 36,\ D_i = 16,\ \rho_{ij}$: 1:1, 4:1, 5:2, 17:1, 19:1, 20:2, 23:1, 40:1, 46:2, 47:2, 52:1, 55:1

$w_i \quad \langle -3003t^3 + 5733t^2 - 966t - 3171,$
$-1365t^3 + 4914t^2 - 6363t - 231\rangle$

$i = 37,\ D_i = 4,\ \rho_{ij}$: 5:1, 7:1, 27:1, 28:2, 51:1, 56:2, 58:2

$w_i \quad \langle 390t^3 - 570t^2 + 270t,$
$300t^3 - 90t^2 - 240t + 210\rangle$

$i = 38,\ D_i = 4,\ \rho_{ij}$: 4:1, 16:1, 21:2, 24:1, 25:2, 28:2, 39:2, 56:1, 58:2

$w_i \quad \langle 405t^3 - 1485t^2 + 1875t - 720,$
$-525t^3 + 30t^2 + 1035t - 885\rangle$

$i = 39,\ D_i = 2,\ \rho_{ij}$: 17:2, 38:1, 52:1, 56:2

$w_i \quad \langle 12t^3 - 28t^2 - 76t - 4,$
$12t^3 + 32t^2 - 60t - 32\rangle$

$i = 40,\ D_i = 16,\ \rho_{ij}$: 2:1, 3:1, 4:1, 6:1, 7:1, 16:1, 20:1, 22:2, 24:2, 28:2, 29:1, 31:1, 32:2, 36:2, 42:1, 48:1, 51:2, 56:2

$w_i \quad \langle -1091960t^3 + 2634700t^2 + 2317480t + 816900,$
$-3852400t^3 - 5432860t^2 - 2820400t - 605220\rangle$

$i = 41,\ D_i = 8,\ \rho_{ij}$: 17:1, 18:1, 27:2, 28:2, 50:2, 58:1

$w_i \quad \langle -240t^3 + 360t - 60,$
$-120t^3 - 480t^2 + 420t - 60\rangle$

$i = 42,\ D_i = 8,\ \rho_{ij}$: 10:2, 11:2, 15:1, 24:1, 25:2, 32:2, 40:2, 47:2, 49:1

$w_i \quad \langle -13168t^3 + 6416t^2 + 12076t + 5284,$
$21160t^3 + 32272t^2 + 16184t - 100\rangle$

$i = 43,\ D_i = 228,\ e_i = 2,\ \rho_{ij}$: 0:1, 6:1, 10:1, 11:1, 14:1, 16:2, 21:1, 27:1, 32:2, 33:1, 34:1, 45:2, 52:2, 57:2

$w_i \quad \langle -35102079t^3 + 3551688t^2 - 4435722t + 4590027,$
$8811666t^3 - 31390254t^2 - 1647540t + 10609380\rangle$

$i = 44,\ D_i = 228,\ e_i = 2,\ \rho_{ij}$: 10:1, 11:1, 12:2, 24:2, 25:2, 30:2, 33:1, 35:1, 50:1, 53:1, 57:1

$w_i \quad \langle 2610630t^3 - 772740t^2 + 273780t - 85050,$
$2047680t^3 - 2211300t^2 + 158760t + 520020\rangle$

$i = 45,\ D_i = 4,\ \rho_{ij}$: 5:2, 29:2, 30:2, 35:1, 43:1, 47:1, 48:1

$w_i \quad \langle -4t^5 - 8t^4 - 16t^3 - 4t^2 - 4t,$
$-8t^5 - 4t^4 - 8t^3 + 4t^2 + 4t\rangle$

$i = 46,\ D_i = 4,\ \rho_{ij}$: 8:2, 13:1, 26:2, 28:2, 30:1, 31:2, 34:1, 36:1, 47:1, 48:2

$w_i \quad \langle 8t^5 + 12t^4 + 8t^3 + 80t^2 + 8t + 36,$
$4t^5 + 32t^4 + 8t^3 + 68t^2 - 12\rangle$

$i = 47,\ D_i = 1,\ \rho_{ij}$: 0:1, 1:1, 3:1, 6:2, 10:2, 14:1, 17:1, 29:2, 30:2, 31:2, 34:2, 36:1, 42:1, 45:2, 46:2, 49:1, 51:1, 56:2

$w_i \quad \langle -2973t^5 - 1848t^4 - 3390t^3 - 1080t^2 - 231t + 1206,$
$-3396t^5 - 1407t^4 - 342t^3 + 1443t^2 - 189t + 1119\rangle$

$i = 48,\ D_i = 177885288,\ \rho_{ij}$: 0:1, 1:1, 3:2, 6:1, 7:1, 21:2, 22:1, 24:1, 28:1, 30:1, 34:1, 40:2, 45:2, 46:1, 51:1, 55:2, 57:2

$w_i \quad \langle 6825458500560t^5 + 404689030200t^4 + 15690371828040t^3 - 10858117979520t^2 + 2405898520200t$
$+10021167699480,$
$30618505197000t^5 - 23602709438280t^4 + 15891382203480t^3 - 217909477800t^2 + 3805855736760t$
$+764017311960\rangle$

$i = 49,\ D_i = 1,\ \rho_{ij}$: 2:1, 3:1, 5:2, 12:1, 18:2, 23:2, 25:2, 26:1, 28:2, 33:1, 34:2, 35:2, 42:2, 47:2, 54:1

$w_i \quad \langle 2826t^5 + 882t^4 + 1182t^3 - 540t^2 + 369t - 1041,$
$1923t^5 + 846t^4 + 2700t^3 + 1293t^2 + 225t - 720\rangle$

$i = 50,\ D_i = 137500,\ \rho_{ij}$: 9:1, 17:1, 19:2, 22:1, 23:1, 26:2, 28:2, 29:1, 30:1, 41:1, 44:2, 52:1, 54:1, 56:1, 58:1

$w_i \quad \langle 119437500t^5 - 316312500t^4 + 121500000t^3 + 7312500t^2 - 375000t - 54187500,$
$398250000t^5 - 921187500t^4 + 321937500t^3 + 31875000t^2 + 6562500t - 154125000\rangle$

$i = 51,\ D_i = 1,\ \rho_{ij}$: 1:2, 13:2, 14:2, 16:1, 17:1, 23:2, 28:1, 30:2, 37:2, 40:1, 47:2, 48:2

$w_i \quad \langle 266t^5 + 216t^4 - 61t^3 + 11t^2 + 53t - 137,$
$-296t^5 - 168t^4 + 118t^3 + 46t^2 + 7t - 214\rangle$

$i = 52,\ D_i = 17432758224,\ \rho_{ij}$: 9:2, 13:2, 18:2, 19:2, 20:2, 21:1, 22:1, 24:1, 27:2, 30:2, 35:1, 36:2, 39:2, 43:1, 50:2, 54:2, 56:1

$w_i \quad \langle -27372393980578080t^5 + 121698085161744t^4 + 86232236818784112t^3 + 23845014816017472t^2$

$-451907822492511184t - 11539980394299504,$
$-26321617045868256t^5 - 55699283772194832t^4 + 36318728606310864t^3 + 63143019235568160t^2$
$+15678970448390928t - 33510171794358672\rangle$

$i = 53,\ D_i = 12,\ \rho_{ij}$: 1:2, 2:2, 14:2, 16:2, 35:1, 44:2

$w_i \quad \langle -324t^5 - 540t^4 - 504t^3 - 432t^2 - 72t - 36,$
$-612t^4 - 900t^3 - 684t^2 - 216t - 324\rangle$

$i = 54,\ D_i = 1,\ \rho_{ij}$: 0:2, 1:2, 2:2, 3:2, 6:1, 7:2, 8:1, 12:1, 14:1, 18:2, 21:2, 27:2, 29:1, 31:2, 34:2, 35:1, 49:2, 50:2, 52:1, 58:2

$w_i \quad \langle 834618t^5 + 279159t^4 - 2627154t^3 - 1607949t^2 + 477690t + 1086150,$
$1274007t^5 + 1450440t^4 - 565749t^3 - 2779179t^2 - 655491t + 591513\rangle$

$i = 55,\ D_i = 5776,\ \rho_{ij}$: 4:2, 6:2, 8:2, 20:2, 27:1, 36:2, 48:1

$W \quad \langle 7220x^5 - 7220x^4 - 7220x^3 + 4332x - 10108,$
$4332x^5 - 31768x^4 + 11552x^3 - 7220x^2 - 4332x - 11552\rangle$

$A \quad \langle -25592959264x^{11} - 1725266548032x^{10} + 7908224412576x^9 + 12796479632x^8y$
$+2402727234432x^8 + 12796479632x^7y^3 + 109146443920x^7y^2 + 1055333202592x^7y$
$-18798028579408x^7 + 958983238304x^6y^3 + 284533488288x^6y^2 - 4146812134864x^6y$
$+20826646968128x^6 - 289049892864x^5y^4 - 4050462170576x^5y^3 - 2515260981784x^5y^2$
$-2068513295808x^5y + 315395586224x^5 - 96349964288x^4y^5 + 1047053127536x^4y^4$
$-2418534650448x^4y^3 - 4041429361424x^4y^2 + 11705015192800x^4y - 15650847324032x^4$
$+24087491072x^3y^6 - 1200006714964x^3y^5 - 731297334068x^3y^4$
$+16561376117096x^3y^3 + 15057767957380x^3y^2 - 11141293365536x^3y + 13006095684416x^3$
$-768541512016x^2y^6 + 1353415904608x^2y^5 + 4341770265728x^2y^4$
$-14306276045052x^2y^3 - 6919884544528x^2y^2 + 4902557167248x^2y - 6700838922592x^2$
$+240874910720xy^7 + 2422179468176xy^6 + 68497760168xy^5 + 4231484198958xy^4$
$+11974341744530xy^3 - 9324664183318xy^2 + 7741872494370xy + 175056097980x - 192699928576y^8$
$-1044042191152y^7 - 563599749984y^6 + 19039624947184y^5 + 4675059432818y^4$
$-10651261368818y^3 + 25462754813230y^2 + 1314230627966y - 767081464484,$
$-135492137280x^{11} - 22582022880x^{10} + 9258629380800x^9 + 67746068640x^8y$
$-18449512692960x^8 + 67746068640x^7y^3 + 67746068640x^7y^2 + 203990940016x^7y$
$+7241302003520x^7 + 203990940016x^6y^3 - 181408917136x^6y^2 - 5496464368992x^6y$
$+5322582792816x^6 - 96349964288x^5y^4 - 5496464368992x^5y^3 - 3506988153264x^5y^2$
$+10188255989360x^5y - 5975203254048x^5 - 289049892864x^4y^5 + 703053645664x^4y^4$
$+10698609706448x^4y^3 + 6328611912120x^4y^2 - 3750874000368x^4y - 13691480472144x^4$
$-144524946432x^3y^6 - 1097387268008x^3y^5 - 2452675885386x^3y^4$
$+1884634137258x^3y^3 + 3575577804538x^3y^2 - 5782486177782x^3y + 15503712835604x^3$
$+96349964288x^2y^7 - 583745291448x^2y^6 + 133233934992x^2y^5 + 9344817434792x^2y^4$
$-2982708855400x^2y^3 - 11518713504040x^2y^2 + 9300782490176x^2y - 8874358624792x^2$
$+48174982144xy^7 + 3202586446600xy^6 + 3132242819788xy^5 - 12835035470236xy^4$
$+4076672978452xy^3 + 6204183664004xy^2 - 2853579733936xy - 3939413498096x - 429058434720y^7$
$-9394002665328y^6 - 6025536351952y^5 + 12885044576416y^4 - 11631871586452y^3$
$+1982586659472y^2 + 6884520872696y + 1065851086616\rangle$

$B \quad \langle 89575357424x^5y^6 + 1900911106696x^5y^5 + 3548892610172x^5y^4$
$-4480408187332x^5y^3 + 5638699618052x^5y^2 + 9491790633900x^5y - 2295177100024x^5$
$-414003752800x^4y^6 - 3408181898768x^4y^5 + 965974178028x^4y^4$
$+10715498512404x^4y^3 - 1827456688012x^4y^2 - 2377099979788x^4y + 4441933955176x^4$
$-828007505600x^3y^7 - 6517171803168x^3y^6 + 4884867915992x^3y^5$
$+22315178642968x^3y^4 - 12309084304840x^3y^3 - 1225827475336x^3y^2$
$+14381737638176x^3y - 2730919300288x^3 - 385399857152x^2y^8 + 1736557559472x^2y^7$
$+9442801624420x^2y^6 - 2041915561634x^2y^5 - 650053315866x^2y^4$
$+11383958353314x^2y^3 - 5735550602802x^2y^2 - 1607132715620x^2y + 851342262576x^2$
$+168612437504xy^9 + 1556406500628xy^8 + 1599564384914xy^7 + 2027772948380xy^6$
$+14672446751298xy^5 + 1723503992132xy^4 - 4956492809534xy^3 + 15787424541944xy^2$
$-919909196302xy - 429103855380x - 96349964288y^{10} - 618371059864y^9 - 886998088176y^8$
$+6798901565462y^7 + 6289930417004y^6 - 5402959304192y^5 + 12010465031386y^4$
$+10652334878972y^3 - 5057128148098y^2 + 8713667042986y + 866421971836,$
$-324428395376x^5y^6 - 1507270792072x^5y^5 + 4514866788200x^5y^4$
$+6235090325072x^5y^3 + 3811242930040x^5y^2 + 7114690654112x^5y + 2146756855152x^5$
$-89575357424x^4y^6 - 1900911106696x^4y^5 - 3548892610172x^4y^4$
$+4480408187332x^4y^3 - 5638699618052x^4y^2 - 9491790633900x^4y + 2295177100024x^4$
$-179150714848x^3y^7 - 3881849733072x^3y^6 - 6531850118040x^3y^5$
$+13103218776120x^3y^4 - 11481829533336x^3y^3 - 13817563433224x^3y^2$
$+13580828560032x^3y - 6883753307920x^3 - 481749821440x^2y^8 + 350021354640x^2y^7$
$+13399271076956x^2y^6 + 11475415133716x^2y^5 - 6313836693296x^2y^4$
$+17084880877908x^2y^3 + 10220171177800x^2y^2 - 4068924549248x^2y + 5349681220272x^2$
$+24087491072xy^9 + 1636295358764xy^8 + 5142890985176xy^7 - 3381705939280xy^6$
$-3375488288270xy^5 + 6824149485334xy^4 - 3897640901554xy^3 + 1323830773822xy^2$
$+880668302340xy - 202882232192x - 166354235216y^9 - 4685522137700y^8 - 9512681569114y^7$
$-5229616565280y^6 - 9830188328824y^5 - 9569275330342y^4 - 202844019436y^3$
$-7268339262774y^2 + 418613315706y - 705266427292\rangle$

$i = 56,\ D_i = 3888,\ \rho_{ij}$: 7:1, 8:2, 9:1, 18:1, 23:1, 26:2, 37:1, 38:2, 39:1, 40:1, 47:1, 50:2, 52:2

$W \quad \langle 2412504x^5 - 1709748x^4 - 1306368x^3 + 615276x^2 + 1277208x - 1023516,$
$203148x^5 - 468504x^4 - 131220x^3 + 163296x^2 + 96228x - 209952\rangle$

$$
\begin{aligned}
A \quad \langle & -176319369216x^{19} + 1028529653760x^{18}y + 910983407616x^{18} - 2115832430592x^{17}y^2 \\
& -4025958930432x^{17}y - 910983407616x^{17} + 1293008707584x^{16}y^3 + 4760622968832x^{16}y^2 \\
& +2380311484416x^{16}y - 3585160507392x^{16} + 1528101199872x^{15}y^4 + 3879026122752x^{15}y^3 \\
& +1454634796032x^{15}y^2 + 11357906033664x^{15}y + 9286153445376x^{15} - 2762336784384x^{14}y^5 \\
& -13885150325760x^{14}y^4 - 12224809598976x^{14}y^3 - 3379454576640x^{14}y^2 \\
& -17073592252416x^{14}y - 1719113849856x^{14} + 1175462461440x^{13}y^6 \\
& +11137506822144x^{13}y^5 + 18528227048448x^{13}y^4 - 20835072129024x^{13}y^3 \\
& -25625081659392x^{13}y^2 + 455491703808x^{13}y - 16529940864000x^{13} + 352638738432x^{12}y^7 \\
& -852210284544x^{12}y^6 - 12077876791296x^{12}y^5 + 19027798594560x^{12}y^4 \\
& +86859329260032x^{12}y^3 + 18623733373440x^{12}y^2 - 661197634560x^{12}y \\
& +17044205690880x^{12} - 411411861504x^{11}y^8 - 3732093315072x^{11}y^7 \\
& -852210284544x^{11}y^6 + 4363904388096x^{11}y^5 - 73290084470784x^{11}y^4 \\
& -70197148869120x^{11}y^3 + 76544646160896x^{11}y^2 + 13459045183488x^{11}y \\
& -6875445546307591200x^{11} + 88159684608x^{10}y^9 + 2321538361344x^{10}y^8 \\
& +8676382293504x^{10}y^7 - 3298641532416x^{10}y^6 + 15464678008320x^{10}y^5 \\
& +119566572249600x^{10}y^4 - 32185631522304x^{10}y^3 - 126244668358656x^{10}y^2 \\
& +23854541326848x^{10}y + 14160864459432909024x^{10} - 411411861504x^9y^9 \\
& -6185871203328x^9y^8 - 9734298508800x^9y^7 + 13745564158464x^9y^6 \\
& -86976875506176x^9y^5 - 127298911253760x^9y^4 + 194924735988480x^9y^3 \\
& +77330736681984x^9y^2 - 57502154285568x^9y + 1776387609489485088x^9 \\
& -264479053824x^8y^{10} - 191012649984x^8y^9 + 4657770003456x^8y^8 \\
& -22120734196224x^8y^7 - 3004408585036 8x^8y^6 + 143252140847616x^8y^5 \\
& -79226169901056x^8y^4 - 381231862806528x^8y^3 + 33092941609728x^8y^2 \\
& +3437763040089740304x^8y - 18364179946291196544x^8 + 205705930752x^7y^{11} \\
& +2439084607488x^7y^{10} + 5392434041856x^7y^9 + 19736749391616x^7y^8 \\
& +107624608305408x^7y^7 + 36461376225792x^7y^6 - 150565721349888x^7y^5 \\
& +327109163097600x^7y^4 + 3437939980250068752x^7y^3 + 3437466661923368976x^7y^2 \\
& -7080459853084298352x^7y - 1725574493904242544x^7 - 44079842304x^6y^{12} \\
& -1293008707584x^6y^{11} - 6556876542720x^6y^{10} - 9102487435776x^6y^9 \\
& -67820510704896x^6y^8 - 188661725061120x^6y^7 + 146499355897344x^6y^6 \\
& +231432028716672x^6y^5 - 393891960848256x^6y^4 - 7080246212781931632x^6y^3 \\
& -7080128890608319344x^6y^2 - 8882341488204112 80x^6y + 2398013382682271 0448x^6 \\
& +205705930752x^5y^{12} + 3048855759360x^5y^{11} + 3706380073728x^5y^{10} \\
& -6714829310976x^5y^9 + 121384865744640x^5y^8 + 138164592381696x^5y^7 \\
& -472238370563328x^5y^6 - 55494684800640x^5y^5 + 494283801695616x^5y^4 \\
& -8885755435257355 68x^5y^3 - 2607135794986613400x^5y^2 + 9182202328990310976x^5y \\
& -9325362718794083760x^5 + 132239526912x^4y^{13} + 301212255744x^4y^{12} \\
& -602424511488x^4y^{11} + 27884173577472x^4y^{10} + 46048741926912x^4y^9 \\
& -190766537531136x^4y^8 + 21353010276096x^4y^7 + 641846583788544x^4y^6 \\
& -362470379925888x^4y^5 - 3437975229430631184x^4y^4 + 5744877673952244528x^4y^3 \\
& +1272221081493466572 0x^4y^2 - 856141438524014160x^4y - 1056635675168726 2176x^4 \\
& +40406522112x^3y^{14} - 714460777344x^3y^{13} - 1948084141824x^3y^{12} \\
& -21319134100992x^3y^{11} + 55561484808576x^3y^{10} - 592575262999488x^3y^9 \\
& -1751250609591552x^3y^8 + 757799063737845 12x^3y^7 - 1075149406283245056x^3y^6 \\
& -4128864983528312232x^3y^5 + 4021951266582853050x^3y^4 + 7504296946365754422x^3y^3 \\
& -6613809285332381 76x^3y^2 - 8236391270287194126x^3y + 5882003950055058036x^3 \\
& +22039921152x^2y^{15} + 668544274944x^2y^{14} + 3645770290560x^2y^{13} \\
& +4329007846272x^2y^{12} + 38459662410240x^2y^{11} + 163940280168960x^2y^{10} \\
& -83253965491584x^2y^9 - 389348982100800x^2y^8 + 628582224575232x^2y^7 \\
& +3540387188726209080x^2y^6 + 7079455598567027184x^2y^5 + 4428868475774147976x^2y^4 \\
& -6702232063959317772x^2y^3 - 1304118200656196755 2x^2y^2 + 5106844276843243680x^2y \\
& +3737419311034247112x^2 - 102852965376xy^{15} - 1788906933504xy^{14} - 3536795124864xy^{13} \\
& +5288560709760xy^{12} - 70460947678464xy^{11} - 293167525799808xy^{10} + 686151617248512xy^9 \\
& +1661212700690112xy^8 - 5157399803605747 2xy^7 + 1494932406343567 2xy^6 \\
& +3928324737801543660xy^5 - 5267456413262912952xy^4 - 6698771321850895968xy^3 \\
& +6560795290808608092xy^2 + 5497488887818677956xy - 3093676813317249072x - 44079842304y^{16} \\
& -1616260884480y^{15} - 2339904962304y^{14} - 11146077902592y^{13} - 128230506069120y^{12} \\
& +584015327223552y^{11} - 9497888292254976y^{10} + 102064840194543552y^9 \\
& -804769465621601088y^8 - 2820426690173568048y^7 + 9625819891897859 76y^6 \\
& -3375776538768948588y^5 - 7401880951782882390y^4 + 2560192985378712438y^3 \\
& +1169438796766406340y^2 + 430532777493623862y + 140733348851982 7620, \\
& 117546246144x^{18}y + 411411861504x^{18} - 352638738432x^{17}y^2 - 1616260884480x^{17}y \\
& -1733807130624x^{17} + 29386561536x^{16}y^3 + 3702706753536x^{16}y + 1028529653760x^{16} \\
& +940369969152x^{15}y^4 + 7934371614720x^{15}y^3 + 6788295714816x^{15}y^2 \\
& +764050599936x^{15}y + 6229951045632x^{15} - 1028529653760x^{14}y^5 - 12518675214336x^{14}y^4 \\
& -25948333836288x^{14}y^3 - 7420106787840x^{14}y^2 - 4216971580416x^{14}y - 9873884676096x^{14} \\
& -117546246144x^{13}y^6 + 4114118615040x^{13}y^5 + 21011391498240x^{13}y^4 \\
& +6200564484096x^{13}y^3 - 43976989338624x^{13}y^2 - 13459045183488x^{13}y \\
& -5333660918784x^{13} + 793437161472x^{12}y^7 + 5921392149504x^{12}y^6 \\
& +6920535241728x^{12}y^5 + 4114118615040x^{12}y^4 + 81856267158528x^{12}y^3 \\
& +96138136065024x^{12}y^2 + 4672463284224x^{12}y + 17382151148544x^{12} - 470184984576x^{11}y^8 \\
& -5700992937984x^{11}y^7 - 17969882379264x^{11}y^6 - 5774459341824x^{11}y^5
\end{aligned}
$$

$$
\begin{aligned}
&-25625081659392x^{11}y^4 - 116480983288320x^{11}y^3 + 3416187778560x^{11}y^2\\
&+71350571409408x^{11}y - 1624190608426347072x^{11} + 88159684608x^{10}y^9\\
&+1160769180672x^{10}y^8 + 6949921803264x^{10}y^7 - 1836660096000x^{10}y^6\\
&-51698308382208x^{10}y^5 + 17147058656256x^{10}y^4 + 52175840007168x^{10}y^3\\
&-162324019284480x^{10}y^2 - 77213190435840x^{10}y + 5809289598125970336x^{10}\\
&+675890915328x^9y^9 + 2747643503616x^9y^8 + 7309907182080x^9y^7\\
&+59867772489216x^9y^6 + 81246496006656x^9y^5 - 110434698252288x^9y^4\\
&+8889434864640x^9y^3 + 183445610388480x^9y^2 - 46041395286528x^9y\\
&-2613193313380166304x^9 - 573037949952x^8y^{10} - 4415330870784x^8y^9\\
&-9682872026112x^8y^8 - 29783280116736x^8y^7 - 104741051954688x^8y^6\\
&+25441415649792x^8y^5 + 210837559060224x^8y^4 - 181524463928064x^8y^3\\
&-133517842338816x^8y^2 + 812213305951021344x^8y - 7020600442219751904x^8\\
&+235092492288x^7y^{11} + 2541937572864x^7y^{10} + 8140077545472x^7y^9\\
&+6281377528320x^7y^8 + 57729900137472x^7y^7 + 126898519352832x^7y^6\\
&-173696618598912x^7y^5 - 105993654140160x^7y^4 + 812444291671334688x^7y^3\\
&+812043220206171168x^7y^2 - 2904742979565076944x^7y + 3538514754509232960x^7\\
&-44079842304x^6y^{12} - 448145063424x^6y^{11} - 2688870380544x^6y^{10}\\
&+4518183836160x^6y^9 + 15685077219840x^6y^8 - 126773626466304x^6y^7\\
&-79156376817408x^6y^6 + 397328351887872x^6y^5 - 54229225994496x^6y^4\\
&-2904886246399205328x^6y^3 - 2904345577746785232x^6y^2 + 1306607257892157264x^6y\\
&+8599943273894807472x^6 - 337945457664x^5y^{12} - 1395861672960x^5y^{11}\\
&-6141791361024x^5y^{10} - 46952378694144x^5y^9 - 27770300651520x^5y^8\\
&+209008245604608x^5y^7 - 155535723569664x^5y^6 - 531725954412672x^5y^5\\
&+386100848721024x^5y^4 + 1306692218114878032x^5y^3 + 900260957507320512x^5y^2\\
&+3510387657150406128x^5y - 7072857893988703152x^5 + 286518974976x^4y^{13}\\
&+2288478479616x^4y^{12} + 5660586415872x^4y^{11} + 28174365872640x^4y^{10}\\
&+119019247540992x^4y^9 - 12143996554752x^4y^8 - 286476731793792x^4y^7\\
&+415165994740224x^4y^6 + 268795205049600x^4y^5 - 812755096802760096x^4y^4\\
&+2698389846528448464x^4y^3 + 4962752986774982616x^4y^2 - 2175388891261186608x^4y\\
&-2849918124242283120x^4 - 95506324992x^3y^{14} - 955063249920x^3y^{13}\\
&-5199278621760x^3y^{12} - 12090019155264x^3y^{11} + 4434445740672x^3y^{10}\\
&-381831112843776x^3y^9 + 717326299657728x^3y^8 + 8632600380646272x^3y^7\\
&-19606032696234643 2x^3y^6 - 1011888322068116592x^3y^5 + 2055471455937788532x^3y^4\\
&+1552037301657344700x^3y^3 - 2959495617236746482x^3y^2 - 2923651305986705316x^3y\\
&+3878181107993289960x^3 + 22039921152x^2y^{15} + 246112452864x^2y^{14}\\
&+1649320766208x^2y^{13} - 2428064646912x^2y^{12} - 13565571469056x^2y^{11}\\
&+107371149212160x^2y^{10} + 151853220077184x^2y^9 - 468910342469376x^2y^8\\
&-87904388854656x^2y^7 + 1453124291757716520x^2y^6 + 2904008529497578128x^2y^5\\
&+1454857207534237 68x^2y^4 - 3950775340890871032x^2y^3 - 4603073695399368936x^2y^2\\
&+2883135897210425904x^2y + 601343488137244968x^2 + 168972728832xy^{15} + 708950797056xy^{14}\\
&+2585405195136xy^{13} + 29876745708288xy^{12} + 43823458159488xy^{11} - 223103274027264xy^{10}\\
&+111887828952192xy^9 + 319338644742144xy^8 - 5870655899132544xy^7 - 7942867480543369 68xy^6\\
&-563331400376587416xy^5 - 3258993887540601336xy^4 - 1902372018086457756xy^3\\
&+4261351803616562508xy^2 - 279535845259544904xy - 1798527787733674872x - 44079842304y^{16}\\
&-1116689338368y^{15} - 4360231067904y^{14} - 786702741120y^{13} - 45163369723968y^{12}\\
&-198117225369792y^{11} - 1705184370264960y^{10} + 2404374470035545 6y^9 - 222708823724265984y^8\\
&-1295338203221567184y^7 - 2830389697614399 12y^6 - 1483119786879234360y^5\\
&-2943003261707022708y^4 + 1596975057050751648y^3 + 482803107608346786y^2\\
&-123727422704702796y + 616072241059683696\rangle
\end{aligned}
$$

$$
\begin{aligned}
B\quad \langle &-286518974976x^5y^{14} - 1171789141248x^5y^{13} - 2466022288896x^5y^{12}\\
&+14867967550464x^5y^{11} - 300232839741696x^5y^{10} + 1099260953269632x^5y^9\\
&+4114953894795264x^5y^8 - 1521713250532120 32x^5y^7 - 1288657691090291088x^5y^6\\
&+1383691546658362032x^5y^5 + 2678835616850394972x^5y^4 - 2560569112221331500x^5y^3\\
&+499318300993850928x^5y^2 - 426946621126588452x^5y + 1152424553107325832x^5\\
&-286518974976x^4y^{14} - 1171789141248x^4y^{13} - 2466022288896x^4y^{12}\\
&+14867967550464x^4y^{11} - 300232839741696x^4y^{10} + 1099260953269632x^4y^9\\
&+4114953894795264x^4y^8 - 1521713250532120 32x^4y^7 - 1288657691090291088x^4y^6\\
&+1383691546658362032x^4y^5 + 2678835616850394972x^4y^4 - 2560569112221331500x^4y^3\\
&+499318300993850928x^4y^2 - 426946621126588452x^4y + 1152424553107325832x^4\\
&-573037949952x^3y^{15} - 2725603582464x^3y^{14} - 6494430099456x^3y^{13}\\
&+26447905382400x^3y^{12} - 580641722749440x^3y^{11} + 1798211453550336x^3y^{10}\\
&+9695589060616704x^3y^9 - 298856044913363712x^3y^8 - 2780210482251531552x^3y^7\\
&+1049172838529669280x^3y^6 + 7202593295911939320x^3y^5 - 1549357401975469704x^3y^4\\
&-2415455547640740144x^3y^3 - 1881355075947090 00x^3y^2 + 1735586944712533728x^3y\\
&-934895941026214176x^3 - 44079842304x^2y^{16} + 1612587564288x^2y^{15}\\
&+516101486976x^2y^{14} - 16242197448960x^2y^{13} + 249211918812672x^2y^{12}\\
&-2219650322762880x^2y^{11} + 2350637194203590 4x^2y^{10} - 1842107347165363 20x^2y^9\\
&+9344682462351035 52x^2y^8 + 4645006849128334896x^2y^7 + 2423842666278904584x^2y^6\\
&-1793262697235871786x^2y^5 - 5120129453999238054x^2y^4 - 2758816487362459464x^2y^3\\
&+3198944580349343310x^2y^2 - 2273457679209569988x^2y + 851711694692709312x^2\\
&-40406522112xy^{17} - 538141408128xy^{16} - 2417656906368xy^{15} + 3853721028096xy^{14}\\
&-7227529575552xy^{13} - 376305452920512xy^{12} + 2973532473267264xy^{11}
\end{aligned}
$$

$$
\begin{aligned}
&-29343874481671680xy^{10} + 228266831041609536xy^{9} + 280327237099035048xy^{8}\\
&+453618662054214726xy^{7} + 347892422684534712xy^{6} - 483235185054101983 2xy^{5}\\
&+776136098889432 18xy^{4} + 234078275500460406 0xy^{3} - 384551977637857419 0xy^{2}\\
&+209131674811698947 4xy + 178824523290174828x - 22039921152y^{18} - 819150402816y^{17}\\
&-1568507721984y^{16} - 5980777492608y^{15} - 67506033681792y^{14} + 260563634621568y^{13}\\
&-4592174220985536y^{12} + 48514742531125632y^{11} - 375321026383494912y^{10}\\
&-1629684304859849400y^{9} - 6303922647528758 4y^{8} - 157746865370034522 6y^{7}\\
&-6010317196598779956y^{6} + 1065616012534981734y^{5} - 355880384579405772y^{4}\\
&-2723017405741315272y^{3} + 3560539122003901068y^{2} - 491883418358833554y + 461208097926228708,\\
&-44079842304x^{5}y^{14} - 822823723008x^{5}y^{13} + 1046284034688x^{5}y^{12}\\
&+136229160787 20x^{5}y^{11} - 135833530597632x^{5}y^{10} + 408158297505792x^{5}y^{9}\\
&-1048608687057408x^{5}y^{8} - 1666052083120665 6x^{5}y^{7} - 421349526283183776x^{5}y^{6}\\
&+399799874371543200x^{5}y^{5} + 887213982940173720x^{5}y^{4} - 1646404991871433272x^{5}y^{3}\\
&+261960298452151092x^{5}y^{2} + 152074282863974760x^{5}y + 204869856511542 24x^{5}\\
&-44079842304x^{4}y^{14} - 822823723008x^{4}y^{13} + 1046284034688x^{4}y^{12}\\
&+136229160787 20x^{4}y^{11} - 135833530597632x^{4}y^{10} + 408158297505792x^{4}y^{9}\\
&-1048608687057408x^{4}y^{8} - 1666052083120665 6x^{4}y^{7} - 421349526283183776x^{4}y^{6}\\
&+399799874371543200x^{4}y^{5} + 887213982940173720x^{4}y^{4} - 1646404991871433272x^{4}y^{3}\\
&+261960298452151092x^{4}y^{2} + 152074282863974760x^{4}y + 204869856511542 24x^{4}\\
&-88159684608x^{3}y^{15} - 1704420569088x^{3}y^{14} + 995469772032x^{3}y^{13}\\
&+28640877537024x^{3}y^{12} - 253503173090304x^{3}y^{11} + 635205220881408x^{3}y^{10}\\
&-1553006310773760x^{3}y^{9} - 3471918657848985 6x^{3}y^{8} - 864913080341309760x^{3}y^{7}\\
&+2378003803655080 32x^{3}y^{6} + 2307494465042405040x^{3}y^{5} - 2109858006489301584x^{3}y^{4}\\
&-1671286058924275512x^{3}y^{3} + 653428963664150976x^{3}y^{2} + 2437396817876081 28x^{3}y\\
&-218094875317115136x^{3} - 198359290368x^{2}y^{16} - 422431822080x^{2}y^{15}\\
&-1491367997952x^{2}y^{14} - 24789707425728x^{2}y^{13} - 3415065375168x^{2}y^{12}\\
&-251501961854592x^{2}y^{11} + 5479600592198016x^{2}y^{10} - 522615591030723 84x^{2}y^{9}\\
&+366737638869221760x^{2}y^{8} + 1875456728249513472x^{2}y^{7} + 308138507442225648x^{2}y^{6}\\
&-2269229352877771308x^{2}y^{5} - 2155966657981124868x^{2}y^{4} - 701319771498248838x^{2}y^{3}\\
&+1064750524552732860x^{2}y^{2} - 1018561601405282328x^{2}y + 342481372616726784x^{2}\\
&+95506324992xy^{17} + 653850994176xy^{16} + 1988796773952xy^{15} + 13825356835968xy^{14}\\
&+417638819473 92xy^{13} - 7574513214873 6xy^{12} + 340623848500416xy^{11} - 6580937762475264xy^{10}\\
&+650592064657077 12xy^{9} - 1902873008719522 08xy^{8} - 1019437116131517804xy^{7}\\
&-1154529318557190384xy^{6} - 1907425654267765518xy^{5} + 1699316602153209 18xy^{4}\\
&+1030825866320888490xy^{3} - 1596187926164036394xy^{2} + 1111288742184048804xy\\
&-233918648334012456x - 22039921152y^{18} - 569364629760y^{17} - 2453777888256y^{16}\\
&-1368617881536y^{15} - 22849071960960y^{14} - 112484036784960y^{13} - 895361332372416y^{12}\\
&+1159534890290016 0y^{11} - 1052320055263130 88y^{10} - 707263629282276840y^{9}\\
&-423527317767311112y^{8} - 782664400919315052y^{7} - 2469411737556630408y^{6}\\
&+420173251325990406y^{5} + 88249175203282434y^{4} - 1336091020793868816y^{3}\\
&+1712705448099800682y^{2} - 306915049715885100y + 91593813505591152\rangle
\end{aligned}
$$

$i = 57,\ D_i = 12,\ e_i = 0,\ \rho_{ij}$: $0{:}2, 7{:}2, 20{:}2, 25{:}1, 43{:}1, 44{:}2, 48{:}1$

$$
\begin{aligned}
w_i \quad &\langle 972t^5 + 3456t^4 - 972t^3 - 864t^2 + 648t + 1512,\\
&-324t^5 + 756t^4 - 324t^3 + 216t^2 - 648t - 540\rangle
\end{aligned}
$$

$i = 58,\ D_i = 9,\ e_i = 1,\ \rho_{ij}$: $3{:}2, 6{:}1, 15{:}2, 20{:}1, 21{:}2, 28{:}1, 31{:}2, 37{:}1, 38{:}1, 41{:}2, 50{:}2, 54{:}1$

$$
\begin{aligned}
w_i \quad &\langle -18950355t^5 - 63499545t^4 + 1312200t^3 + 25150500t^2 + 20273490t - 50727465,\\
&-41465520t^5 - 63051210t^4 - 31656825t^3 + 43761870t^2 - 14959080t - 40579785\rangle
\end{aligned}
$$

Table 3: The edges (i, j) with a linear endpoint, the linear endpoint k used, and the norm of $R_{ij} = \mathrm{Res}_t(f_k, g_{kl})$.

(i,j)	k	$N(R_{ij})$	(i,j)	k	$N(R_{ij})$
$(0,7)$	0	3^2	$(17,39)$	17	$2^8 \cdot 3$
$(0,8)$	0	3	$(17,41)$	17	$2^4 \cdot 5^2 \cdot 7^2$
$(0,10)$	0	$2^2 \cdot 3^4$	$(17,47)$	17	$3^4 \cdot 7^3$
$(0,24)$	0	3	$(17,50)$	17	$7^3 \cdot 13^2$
$(0,29)$	0	3	$(17,51)$	17	$5^2 \cdot 7^4$
$(0,30)$	0	3	$(18,27)$	18	3^2
$(0,43)$	0	3^8	$(18,34)$	18	$2^2 \cdot 3 \cdot 13$
$(0,47)$	0	3^2	$(18,41)$	18	2^2
$(0,48)$	0	3	$(18,49)$	18	$2^4 \cdot 3^3 \cdot 13$
$(0,54)$	0	3^3	$(18,52)$	18	$2^4 \cdot 3^2$
$(0,57)$	0	3^6	$(18,54)$	18	3^3
$(1,24)$	1	2^2	$(18,56)$	18	$2^4 \cdot 3^8$
$(1,32)$	1	7	$(19,28)$	19	$3^4 \cdot 13 \cdot 19$
$(1,35)$	1	3^2	$(19,36)$	19	$2^2 \cdot 3^4 \cdot 7^4$

(i,j)	k	$N(R_{ij})$	(i,j)	k	$N(R_{ij})$
$(1,36)$	1	$2^2 \cdot 3^4$	$(19,50)$	19	$2^4 \cdot 3^8 \cdot 5^2 \cdot 7^2 \cdot 13$
$(1,47)$	1	3^4	$(19,52)$	19	$2^6 \cdot 3^8 \cdot 5^6 \cdot 19$
$(1,48)$	1	7	$(20,22)$	20	$2^2 \cdot 7$
$(1,51)$	1	7^2	$(20,23)$	20	$2^2 \cdot 7^2$
$(1,53)$	1	3^3	$(20,27)$	20	$2^2 \cdot 3^3$
$(1,54)$	1	3^6	$(20,31)$	20	2^2
$(2,20)$	2	2^2	$(20,33)$	20	$2^2 \cdot 7^2$
$(2,21)$	2	2^2	$(20,35)$	20	$2^2 \cdot 3^3$
$(2,22)$	2	1	$(20,36)$	20	$2^2 \cdot 7^2$
$(2,23)$	2	1	$(20,40)$	20	2^4
$(2,24)$	2	$2^2 \cdot 7 \cdot 13$	$(20,52)$	20	$2^4 \cdot 5^2 \cdot 7$
$(2,33)$	2	1	$(20,55)$	20	$3^2 \cdot 5^2 \cdot 7^2$
$(2,40)$	2	$2^2 \cdot 7 \cdot 19$	$(20,57)$	20	$2^4 \cdot 3^4 \cdot 5^2 \cdot 7$
$(2,49)$	2	$2^4 \cdot 19$	$(20,58)$	20	$3^6 \cdot 7^4$
$(2,53)$	2	$13^2 \cdot 19$	$(21,22)$	21	2^4
$(2,54)$	2	19	$(21,25)$	21	$2^2 \cdot 19$
$(3,5)$	3	1	$(21,32)$	21	2^8
$(3,6)$	3	1	$(21,34)$	21	2^2
$(3,7)$	3	2^2	$(21,38)$	21	$2^2 \cdot 19$
$(3,10)$	3	$2^2 \cdot 13^2$	$(21,43)$	21	$2^8 \cdot 3^4$
$(3,14)$	3	13	$(21,48)$	21	2^2
$(3,17)$	3	$5^2 \cdot 7$	$(21,52)$	21	2^6
$(3,18)$	3	13	$(21,54)$	21	2^2
$(3,19)$	3	$3^4 \cdot 7$	$(21,58)$	21	$3^4 \cdot 19$
$(3,22)$	3	$2^2 \cdot 7 \cdot 13$	$(22,26)$	22	7^2
$(3,26)$	3	$5^2 \cdot 7^2$	$(22,40)$	22	$2^2 \cdot 3 \cdot 7 \cdot 13$
$(3,40)$	3	2^{12}	$(22,48)$	22	$7^2 \cdot 13$
$(3,47)$	3	$7 \cdot 13$	$(22,50)$	22	$3^4 \cdot 7$
$(3,48)$	3	$5^2 \cdot 13$	$(22,52)$	22	$2^6 \cdot 3$
$(3,49)$	3	$7 \cdot 13$	$(23,25)$	23	7^2
$(3,54)$	3	7	$(23,26)$	23	$2^2 \cdot 3^2 \cdot 7$
$(3,58)$	3	3^4	$(23,36)$	23	$3^4 \cdot 7$
$(4,6)$	4	3^2	$(23,49)$	23	7^3
$(4,8)$	4	2^2	$(23,50)$	23	$3^2 \cdot 7^2$
$(4,18)$	4	3^5	$(23,51)$	23	$3 \cdot 7^2$
$(4,19)$	4	$5^2 \cdot 7$	$(23,56)$	23	$3^4 \cdot 7^2$
$(4,22)$	4	7	$(24,28)$	24	3^4
$(4,27)$	4	3^2	$(24,35)$	24	$13 \cdot 19$
$(4,36)$	4	$3^2 \cdot 7^2$	$(24,38)$	24	$5^2 \cdot 13$
$(4,38)$	4	3^2	$(24,40)$	24	$3 \cdot 5^2 \cdot 7 \cdot 19$
$(4,40)$	4	$2^2 \cdot 3 \cdot 5^2 \cdot 7$	$(24,42)$	24	5^6
$(4,55)$	4	$3^3 \cdot 7$	$(24,44)$	24	$2^2 \cdot 3^5$
$(5,14)$	5	$2^2 \cdot 3^2 \cdot 13$	$(24,48)$	24	$5^2 \cdot 13$
$(5,17)$	5	$3^2 \cdot 19$	$(24,52)$	24	$2^2 \cdot 3 \cdot 5^2 \cdot 13$
$(5,22)$	5	3	$(25,26)$	25	3^2
$(5,26)$	5	13	$(25,27)$	25	3^4
$(5,27)$	5	3^2	$(25,28)$	25	$2^2 \cdot 7 \cdot 13$
$(5,29)$	5	$3^2 \cdot 7$	$(25,30)$	25	$2^2 \cdot 7^2$
$(5,31)$	5	$2^2 \cdot 3$	$(25,35)$	25	3^6
$(5,32)$	5	3^2	$(25,38)$	25	$3^2 \cdot 13 \cdot 19$
$(5,36)$	5	$7 \cdot 19$	$(25,42)$	25	$2^2 \cdot 3 \cdot 7$
$(5,37)$	5	3^2	$(25,44)$	25	$2^2 \cdot 3^5 \cdot 19$
$(5,45)$	5	3^2	$(25,49)$	25	$3^4 \cdot 7$
$(5,49)$	5	3^2	$(25,57)$	25	$3^4 \cdot 7^2 \cdot 19$
$(6,12)$	6	$3 \cdot 7$	$(26,27)$	26	$2^2 \cdot 3^2 \cdot 5^2$
$(6,16)$	6	3^2	$(26,46)$	26	$3^4 \cdot 7$
$(6,18)$	6	$2^2 \cdot 3^2 \cdot 13$	$(26,49)$	26	$3^4 \cdot 7$

(i,j)	k	$N(R_{ij})$	(i,j)	k	$N(R_{ij})$
$(6,27)$	6	3^4	$(26,50)$	26	$5^4 \cdot 7^2$
$(6,32)$	6	3^2	$(26,56)$	26	$3^6 \cdot 5^2 \cdot 7^2$
$(6,34)$	6	13	$(27,28)$	27	$13 \cdot 19$
$(6,40)$	6	$3 \cdot 13$	$(27,37)$	27	$2^6 \cdot 3^2 \cdot 13$
$(6,43)$	6	$2^2 \cdot 3^6$	$(27,41)$	27	1
$(6,47)$	6	3^2	$(27,43)$	27	$2^4 \cdot 3^5$
$(6,48)$	6	$7 \cdot 13$	$(27,52)$	27	19
$(6,54)$	6	3^4	$(27,54)$	27	3^5
$(6,55)$	6	$3^2 \cdot 7$	$(27,55)$	27	3^3
$(6,58)$	6	$3^3 \cdot 7 \cdot 13$	$(28,35)$	28	19^2
$(7,14)$	7	3^2	$(28,37)$	28	$5^2 \cdot 13$
$(7,16)$	7	$2^2 \cdot 3^2$	$(28,38)$	28	13^2
$(7,25)$	7	$3^2 \cdot 7$	$(28,40)$	28	$3 \cdot 5^2 \cdot 7 \cdot 19$
$(7,29)$	7	$3 \cdot 7$	$(28,41)$	28	$3^5 \cdot 7$
$(7,30)$	7	$2^2 \cdot 3$	$(28,46)$	28	$3^3 \cdot 19$
$(7,37)$	7	3^2	$(28,48)$	28	$2^2 \cdot 5^2 \cdot 7^2$
$(7,40)$	7	$2^2 \cdot 3^2 \cdot 7^2$	$(28,49)$	28	$7 \cdot 19$
$(7,48)$	7	7	$(28,50)$	28	$3^4 \cdot 13$
$(7,54)$	7	3^2	$(28,51)$	28	$3^2 \cdot 5^2$
$(7,56)$	7	$3^2 \cdot 7^2$	$(28,58)$	28	$3^6 \cdot 7$
$(7,57)$	7	$2^2 \cdot 3^3 \cdot 7^2$	$(29,32)$	29	$2^2 \cdot 3^2 \cdot 7$
$(8,12)$	8	3^2	$(29,33)$	29	$3^3 \cdot 7^2$
$(8,15)$	8	7^2	$(29,40)$	29	$3^4 \cdot 7^2$
$(8,17)$	8	7	$(29,45)$	29	$3^2 \cdot 7^2$
$(8,19)$	8	5^2	$(29,47)$	29	7^3
$(8,23)$	8	7^2	$(29,50)$	29	$3^2 \cdot 7^2$
$(8,29)$	8	7	$(29,54)$	29	$3 \cdot 7^3$
$(8,32)$	8	7	$(30,31)$	30	$2^2 \cdot 3 \cdot 7$
$(8,46)$	8	3	$(30,44)$	30	$2^2 \cdot 3^5$
$(8,54)$	8	$2^2 \cdot 3 \cdot 7^2$	$(30,45)$	30	7
$(8,55)$	8	5^2	$(30,46)$	30	$3^4 \cdot 7$
$(8,56)$	8	$2^6 \cdot 7$	$(30,47)$	30	7^2
$(9,17)$	9	$3^2 \cdot 7^3$	$(30,48)$	30	13
$(9,29)$	9	3^2	$(30,50)$	30	$3^4 \cdot 7$
$(9,31)$	9	$2^2 \cdot 3^4$	$(30,51)$	30	$2^4 \cdot 3^2 \cdot 7$
$(9,33)$	9	2^8	$(30,52)$	30	$2^4 \cdot 3 \cdot 7$
$(9,50)$	9	$5^2 \cdot 13$	$(31,40)$	31	$3^5 \cdot 13$
$(9,52)$	9	$3^2 \cdot 5^2 \cdot 13$	$(31,46)$	31	$2^4 \cdot 3^2 \cdot 7$
$(9,56)$	9	$2^8 \cdot 3^4$	$(31,47)$	31	$2^4 \cdot 7^2$
$(10,12)$	12	$2^4 \cdot 3 \cdot 13$	$(31,54)$	31	$2^2 \cdot 3$
$(10,14)$	14	$2^4 \cdot 3^2 \cdot 13$	$(31,58)$	31	$3^4 \cdot 7^2 \cdot 13$
$(10,17)$	17	$2^4 \cdot 3^2 \cdot 7$	$(32,40)$	32	$3^2 \cdot 7^2$
$(10,20)$	20	$2^6 \cdot 3^2$	$(32,42)$	32	7^2
$(10,22)$	22	$2^4 \cdot 3 \cdot 13$	$(32,43)$	32	$2^2 \cdot 3^6 \cdot 7$
$(10,24)$	24	$2^4 \cdot 3 \cdot 19$	$(33,43)$	33	$2^4 \cdot 3^6 \cdot 7$
$(10,29)$	29	$2^4 \cdot 3 \cdot 7$	$(33,44)$	33	$2^2 \cdot 3^7$
$(10,30)$	30	$2^4 \cdot 3 \cdot 13$	$(33,49)$	33	7^3
$(10,32)$	32	$2^4 \cdot 3^2$	$(34,35)$	34	3
$(10,33)$	33	$2^6 \cdot 3^2 \cdot 7$	$(34,43)$	34	$2^4 \cdot 3^5$
$(10,35)$	35	$2^4 \cdot 3^2 \cdot 19$	$(34,46)$	34	$3^2 \cdot 13$
$(10,42)$	42	$2^2 \cdot 13^2$	$(34,47)$	34	$3^4 \cdot 7$
$(10,43)$	43	$2^4 \cdot 3^4 \cdot 7$	$(34,48)$	34	$3 \cdot 7^2 \cdot 13$
$(10,44)$	44	$2^4 \cdot 3^4 \cdot 7$	$(34,49)$	34	3^3
$(10,47)$	47	$3^2 \cdot 7^2 \cdot 13$	$(34,54)$	34	$3^2 \cdot 7$
$(11,15)$	15	$2^4 \cdot 7 \cdot 19$	$(35,44)$	35	$2^4 \cdot 3^5$
$(11,20)$	20	$2^4 \cdot 7^2$	$(35,45)$	35	3^2
$(11,22)$	22	$2^4 \cdot 7^2$	$(35,49)$	35	$3^5 \cdot 19$

(i,j)	k	$N(R_{ij})$	(i,j)	k	$N(R_{ij})$
$(11,24)$	24	$2^4 \cdot 5^2$	$(35,52)$	35	$13 \cdot 19$
$(11,30)$	30	$2^4 \cdot 7^2$	$(35,53)$	35	$3^3 \cdot 19$
$(11,31)$	31	2^8	$(35,54)$	35	$3^4 \cdot 19$
$(11,34)$	34	$2^4 \cdot 3 \cdot 7$	$(36,40)$	36	$3^2 \cdot 7^4$
$(11,42)$	42	$2^2 \cdot 3 \cdot 5^2 \cdot 19$	$(36,46)$	36	3^3
$(11,43)$	43	$2^4 \cdot 3^4$	$(36,47)$	36	$3^5 \cdot 7$
$(11,44)$	44	$2^4 \cdot 3^4$	$(36,52)$	36	$2^2 \cdot 3^4$
$(12,17)$	12	$7^3 \cdot 13$	$(36,55)$	36	$3^2 \cdot 7^3$
$(12,23)$	12	$3 \cdot 7$	$(37,51)$	37	5^6
$(12,29)$	12	$3^2 \cdot 7$	$(37,56)$	37	$3^2 \cdot 5^2 \cdot 7^3$
$(12,33)$	12	$2^2 \cdot 3^3$	$(37,58)$	37	$3^6 \cdot 5^4$
$(12,34)$	12	$3 \cdot 13$	$(38,39)$	38	$5^2 \cdot 13$
$(12,44)$	12	3^{12}	$(38,56)$	38	$3^2 \cdot 5^2 \cdot 7^3$
$(12,49)$	12	$3^2 \cdot 7^2 \cdot 13$	$(38,58)$	38	$3^6 \cdot 5^4 \cdot 19$
$(12,54)$	12	$3 \cdot 7^2$	$(39,52)$	39	$2^4 \cdot 3^4 \cdot 13^2 \cdot 19$
$(13,21)$	13	$2^2 \cdot 3^4$	$(39,56)$	39	$2^4 \cdot 3^6 \cdot 5^4 \cdot 19$
$(13,46)$	13	$2^4 \cdot 7 \cdot 19$	$(40,42)$	40	$2^2 \cdot 3 \cdot 5^2 \cdot 7$
$(13,51)$	13	$5^4 \cdot 7$	$(40,48)$	40	$3^2 \cdot 5^2 \cdot 7 \cdot 13^2$
$(13,52)$	13	$5^4 \cdot 19$	$(40,51)$	40	$3 \cdot 5^2 \cdot 7^2$
$(14,16)$	14	$2^2 \cdot 3^6 \cdot 7$	$(40,56)$	40	$3^6 \cdot 5^4 \cdot 7^2$
$(14,22)$	14	13^2	$(41,50)$	41	$3^4 \cdot 5^2$
$(14,24)$	14	$2^4 \cdot 5^2$	$(41,58)$	41	$3^8 \cdot 5^2 \cdot 7^2$
$(14,26)$	14	$3^2 \cdot 5^2 \cdot 13$	$(42,47)$	42	$2^4 \cdot 3^3 \cdot 13^2$
$(14,27)$	14	$3^2 \cdot 5^2$	$(42,49)$	42	$3^4 \cdot 13$
$(14,29)$	14	3	$(43,45)$	43	$3^{12} \cdot 7^2$
$(14,43)$	14	$2^4 \cdot 3^4 \cdot 7$	$(43,52)$	43	$2^4 \cdot 3^{14}$
$(14,47)$	14	$2^4 \cdot 3^3 \cdot 13$	$(43,57)$	43	$2^{10} \cdot 3^{18}$
$(14,51)$	14	7	$(44,50)$	44	$3^{12} \cdot 5^2$
$(14,53)$	14	$3^3 \cdot 5^2 \cdot 7$	$(44,53)$	44	$3^{10} \cdot 5^4$
$(14,54)$	14	$2^2 \cdot 3^4$	$(44,57)$	44	$2^4 \cdot 3^{16} \cdot 5^2 \cdot 19$
$(15,28)$	15	1	$(45,47)$	45	$3^2 \cdot 7$
$(15,42)$	15	$5^2 \cdot 7 \cdot 19$	$(45,48)$	45	3^4
$(15,58)$	15	$3^4 \cdot 5^2 \cdot 7^2$	$(46,47)$	46	$3^2 \cdot 7$
$(16,24)$	16	$2^2 \cdot 13 \cdot 19$	$(46,48)$	46	$3 \cdot 7 \cdot 13$
$(16,25)$	16	$2^2 \cdot 3^2 \cdot 7^2$	$(47,49)$	47	3^8
$(16,30)$	16	$2^8 \cdot 7$	$(47,51)$	47	7^3
$(16,34)$	16	$2^4 \cdot 3 \cdot 13$	$(47,56)$	47	$2^4 \cdot 3^2 \cdot 7^2$
$(16,38)$	16	3^2	$(48,51)$	48	$5^2 \cdot 7^2$
$(16,40)$	16	$2^2 \cdot 3 \cdot 7 \cdot 19$	$(48,55)$	48	$5^4 \cdot 7$
$(16,43)$	16	$2^6 \cdot 3^4 \cdot 7^2$	$(48,57)$	48	$3^6 \cdot 5^4 \cdot 7$
$(16,51)$	16	$2^4 \cdot 7^3$	$(49,54)$	49	$3^5 \cdot 7 \cdot 19$
$(16,53)$	16	$3^3 \cdot 7^2 \cdot 13$	$(50,52)$	50	$3^2 \cdot 13^2$
$(17,22)$	17	$2^4 \cdot 7^2$	$(50,54)$	50	7^3
$(17,27)$	17	$3^3 \cdot 13$	$(50,56)$	50	7^2
$(17,28)$	17	$2^2 \cdot 7 \cdot 13$	$(50,58)$	50	$3^6 \cdot 5^4 \cdot 7^2$
$(17,32)$	17	1	$(52,54)$	52	$3^2 \cdot 7 \cdot 19$
$(17,34)$	17	$2^4 \cdot 3 \cdot 13$	$(52,56)$	52	$2^8 \cdot 3^6 \cdot 19$
$(17,36)$	17	$2^2 \cdot 3^2 \cdot 7 \cdot 19$	$(54,58)$	54	$3^8 \cdot 7^2$

Table 4: The terminal residue classes $x \equiv a$, $y \equiv b \pmod{p^k}$ of Step 4, grouped by p, k and the value of B_p.

p	k	B_p	classes (a,b)
2	2	1	$(3,3)$
2	3	1	$(3,1)$, $(3,5)$

p	k	B_p	classes (a, b)
3	3	0	$(1,15)$, $(4,0)$, $(7,3)$, $(10,24)$, $(13,9)$, $(16,12)$, $(19,6)$, $(22,18)$, $(25,21)$
3	3	2	$(0,4)$, $(2,6)$, $(3,16)$, $(6,10)$, $(8,21)$, $(9,13)$, $(11,15)$, $(17,3)$, $(18,22)$, $(20,24)$, $(26,12)$
3	4	2	$(5,9)$, $(12,52)$, $(14,72)$, $(15,73)$, $(21,61)$, $(23,54)$, $(24,28)$, $(32,36)$, $(39,79)$, $(41,18)$, $(42,19)$, $(48,7)$, $(50,0)$, $(51,55)$, $(59,63)$, $(66,25)$, $(68,45)$, $(69,46)$, $(75,34)$, $(77,27)$, $(78,1)$
5	1	0	$(3,4)$
5	2	0	$(2,3)$, $(2,8)$, $(2,13)$, $(2,18)$, $(2,23)$
7	2	1	$(4,11)$, $(5,10)$, $(5,18)$, $(8,24)$, $(12,4)$, $(12,6)$, $(12,38)$, $(18,18)$, $(19,17)$, $(19,39)$, $(22,31)$, $(26,45)$, $(30,30)$, $(31,29)$, $(33,24)$, $(36,38)$, $(37,37)$, $(40,3)$, $(40,48)$, $(43,17)$, $(45,29)$, $(47,31)$, $(47,32)$, $(47,34)$
7	3	1	$(1,290)$, $(2,247)$, $(3,274)$, $(5,167)$, $(9,107)$, $(10,127)$, $(11,39)$, $(15,150)$, $(16,310)$, $(17,29)$, $(19,41)$, $(23,170)$, $(24,323)$, $(25,46)$, $(26,76)$, $(26,172)$, $(29,59)$, $(32,123)$, $(33,60)$, $(33,62)$, $(38,29)$, $(39,249)$, $(40,291)$, $(44,93)$, $(46,81)$, $(50,143)$, $(51,296)$, $(52,274)$, $(54,69)$, $(58,156)$, $(59,127)$, $(60,235)$, $(64,3)$, $(65,16)$, $(66,29)$, $(68,286)$, $(72,219)$, $(73,323)$, $(74,242)$, $(75,74)$, $(75,321)$, $(78,255)$, $(82,305)$, $(82,307)$, $(87,29)$, $(88,102)$, $(89,193)$, $(93,142)$, $(95,277)$, $(99,339)$, $(100,2)$, $(101,274)$, $(103,314)$, $(107,205)$, $(108,127)$, $(109,88)$, $(113,199)$, $(114,65)$, $(115,29)$, $(117,188)$, $(121,268)$, $(122,323)$, $(123,95)$, $(124,223)$, $(124,319)$, $(127,108)$, $(131,207)$, $(131,209)$, $(136,29)$, $(137,298)$, $(138,95)$, $(142,191)$, $(144,130)$, $(148,192)$, $(149,51)$, $(150,274)$, $(152,216)$, $(156,254)$, $(157,127)$, $(158,284)$, $(162,52)$, $(163,114)$, $(164,29)$, $(166,90)$, $(170,317)$, $(171,323)$, $(172,291)$, $(173,125)$, $(173,221)$, $(176,304)$, $(179,25)$, $(180,109)$, $(180,111)$, $(185,29)$, $(186,151)$, $(187,340)$, $(191,240)$, $(193,326)$, $(197,45)$, $(198,100)$, $(199,274)$, $(201,118)$, $(205,303)$, $(206,127)$, $(207,137)$, $(211,248)$, $(212,163)$, $(213,29)$, $(215,335)$, $(219,23)$, $(220,323)$, $(221,144)$, $(222,27)$, $(222,123)$, $(225,157)$, $(229,11)$, $(229,13)$, $(234,29)$, $(235,4)$, $(236,242)$, $(240,289)$, $(242,179)$, $(246,241)$, $(247,149)$, $(248,274)$, $(250,20)$, $(254,9)$, $(255,127)$, $(256,333)$, $(260,101)$, $(261,212)$, $(262,29)$, $(264,237)$, $(268,72)$, $(269,323)$, $(270,340)$, $(271,25)$, $(271,272)$, $(274,10)$, $(278,256)$, $(283,29)$, $(284,200)$, $(285,144)$, $(289,338)$, $(291,32)$, $(295,94)$, $(296,198)$, $(297,274)$, $(299,265)$, $(303,58)$, $(304,127)$, $(305,186)$, $(309,297)$, $(310,261)$, $(311,29)$, $(313,139)$, $(317,121)$, $(318,323)$, $(319,193)$, $(320,174)$, $(320,270)$, $(323,206)$, $(327,158)$, $(327,160)$, $(332,29)$, $(333,53)$, $(334,46)$, $(338,44)$, $(340,228)$
7	4	1	$(81,2377)$, $(130,1887)$, $(228,907)$, $(277,417)$, $(278,1287)$, $(326,2328)$, $(424,1348)$, $(473,858)$, $(571,2279)$, $(620,1789)$, $(621,601)$, $(669,1299)$, $(767,319)$, $(816,2230)$, $(914,1250)$, $(963,760)$, $(964,2316)$, $(1012,270)$, $(1110,1691)$, $(1159,1201)$, $(1257,221)$, $(1306,2132)$, $(1307,1630)$, $(1355,1642)$, $(1453,662)$, $(1502,172)$, $(1600,1593)$, $(1649,1103)$, $(1650,944)$, $(1698,613)$, $(1796,2034)$, $(1845,1544)$, $(1943,564)$, $(1992,74)$, $(1993,258)$, $(2041,1985)$, $(2139,1005)$, $(2188,515)$, $(2286,1936)$, $(2335,1446)$, $(2336,1973)$, $(2384,956)$
11	1	0	$(0,6)$, $(2,0)$, $(2,4)$, $(2,6)$, $(4,3)$, $(7,2)$, $(7,9)$, $(7,10)$, $(8,4)$, $(8,8)$, $(8,9)$, $(9,0)$, $(9,1)$, $(9,9)$
13	1	0	$(0,2)$, $(3,3)$, $(4,2)$, $(5,9)$, $(8,1)$, $(9,9)$, $(12,2)$

p	k	B_p	classes (a, b)
13	2	0	$(6,64)$, $(9,23)$, $(9,84)$, $(10,21)$, $(11,149)$, $(19,116)$, $(21,49)$, $(22,49)$, $(22,97)$, $(23,21)$, $(24,58)$, $(32,168)$, $(34,114)$, $(35,75)$, $(35,110)$, $(36,21)$, $(37,136)$, $(45,51)$, $(47,10)$, $(48,101)$, $(48,123)$, $(49,21)$, $(50,45)$, $(58,103)$, $(60,75)$, $(61,127)$, $(61,136)$, $(62,21)$, $(63,123)$, $(71,155)$, $(73,140)$, $(74,149)$, $(74,153)$, $(75,21)$, $(76,32)$, $(84,38)$, $(86,36)$, $(87,10)$, $(87,162)$, $(88,21)$, $(89,110)$, $(99,101)$, $(100,6)$, $(100,36)$, $(101,21)$, $(102,19)$, $(110,142)$, $(112,166)$, $(113,19)$, $(113,62)$, $(114,21)$, $(125,62)$, $(126,32)$, $(126,88)$, $(127,21)$, $(128,6)$, $(136,77)$, $(138,127)$, $(139,45)$, $(139,114)$, $(140,21)$, $(141,84)$, $(149,129)$, $(152,58)$, $(152,140)$, $(153,21)$, $(154,162)$, $(162,12)$, $(164,88)$, $(165,71)$, $(165,166)$, $(166,2)$, $(166,15)$, $(166,21)$, $(166,28)$, $(166,41)$, $(166,54)$, $(166,67)$, $(166,80)$, $(166,93)$, $(166,106)$, $(166,119)$, $(166,132)$, $(166,145)$, $(166,158)$, $(167,71)$
13	3	0	$(8,1505)$, $(97,2118)$, $(115,1280)$, $(123,1377)$, $(151,23)$, $(177,153)$, $(266,597)$, $(284,97)$, $(292,2053)$, $(320,868)$, $(346,998)$, $(435,1273)$, $(453,1111)$, $(461,532)$, $(489,1713)$, $(515,1843)$, $(604,1949)$, $(622,2125)$, $(630,1208)$, $(658,361)$, $(684,491)$, $(773,428)$, $(791,942)$, $(799,1884)$, $(827,1206)$, $(853,1336)$, $(942,1104)$, $(960,1956)$, $(968,363)$, $(996,2051)$, $(1022,2181)$, $(1111,1780)$, $(1129,773)$, $(1137,1039)$, $(1165,699)$, $(1191,829)$, $(1280,259)$, $(1298,1787)$, $(1306,1715)$, $(1334,1544)$, $(1360,1674)$, $(1449,935)$, $(1467,604)$, $(1475,194)$, $(1503,192)$, $(1529,322)$, $(1618,1611)$, $(1636,1618)$, $(1644,870)$, $(1672,1037)$, $(1698,1167)$, $(1787,90)$, $(1805,435)$, $(1813,1546)$, $(1841,1882)$, $(1867,2012)$, $(1956,766)$, $(1974,1449)$, $(1982,25)$, $(2010,530)$, $(2036,660)$, $(2125,1442)$, $(2143,266)$, $(2151,701)$, $(2179,1375)$
19	1	0	$(3,12)$, $(4,0)$, $(4,10)$, $(5,2)$, $(7,7)$, $(8,6)$, $(8,15)$, $(8,16)$, $(10,7)$, $(11,5)$, $(11,11)$, $(12,10)$, $(14,4)$, $(15,6)$, $(17,12)$, $(17,14)$, $(18,11)$, $(18,13)$
19	2	0	$(4,236)$, $(9,136)$, $(10,175)$, $(11,173)$, $(15,133)$, $(15,297)$, $(16,288)$, $(17,220)$, $(23,312)$, $(28,307)$, $(29,4)$, $(30,59)$, $(34,88)$, $(34,228)$, $(35,98)$, $(36,315)$, $(42,27)$, $(47,117)$, $(48,194)$, $(49,306)$, $(53,240)$, $(53,323)$, $(54,269)$, $(55,49)$, $(61,103)$, $(66,288)$, $(67,23)$, $(68,192)$, $(72,31)$, $(72,57)$, $(73,79)$, $(74,144)$, $(80,179)$, $(85,98)$, $(86,213)$, $(87,78)$, $(91,152)$, $(91,183)$, $(92,250)$, $(93,239)$, $(99,255)$, $(104,269)$, $(105,42)$, $(106,325)$, $(110,247)$, $(110,335)$, $(111,60)$, $(112,334)$, $(118,331)$, $(123,79)$, $(124,232)$, $(125,211)$, $(129,126)$, $(129,342)$, $(130,231)$, $(131,68)$, $(137,46)$, $(142,250)$, $(143,61)$, $(144,97)$, $(148,76)$, $(148,278)$, $(149,41)$, $(150,163)$, $(156,122)$, $(161,60)$, $(162,251)$, $(163,344)$, $(167,69)$, $(167,171)$, $(168,212)$, $(169,258)$, $(175,198)$, $(180,231)$, $(181,80)$, $(182,230)$, $(186,221)$, $(186,266)$, $(187,22)$, $(188,353)$, $(194,274)$, $(199,41)$, $(200,270)$, $(201,116)$, $(205,0)$, $(205,12)$, $(206,193)$, $(207,87)$, $(213,350)$, $(218,212)$, $(219,99)$, $(220,2)$, $(224,95)$, $(224,164)$, $(225,3)$, $(226,182)$, $(232,65)$, $(237,22)$, $(238,289)$, $(239,249)$, $(243,190)$, $(243,316)$, $(244,174)$, $(245,277)$, $(251,141)$, $(256,193)$, $(257,118)$, $(258,135)$, $(262,107)$, $(262,285)$, $(263,345)$, $(264,11)$, $(270,217)$, $(275,3)$, $(276,308)$, $(277,21)$, $(281,19)$, $(281,259)$, $(282,155)$, $(283,106)$, $(289,293)$, $(294,174)$, $(295,137)$, $(296,268)$, $(300,50)$, $(300,114)$, $(301,326)$, $(302,201)$, $(308,8)$, $(313,345)$, $(314,327)$, $(315,154)$, $(319,202)$, $(319,209)$, $(320,136)$, $(321,296)$, $(327,84)$, $(332,155)$, $(333,156)$, $(334,40)$, $(338,304)$, $(338,354)$, $(339,307)$, $(340,30)$, $(346,160)$, $(351,326)$, $(352,346)$, $(353,287)$, $(357,38)$, $(357,145)$, $(358,117)$, $(359,125)$